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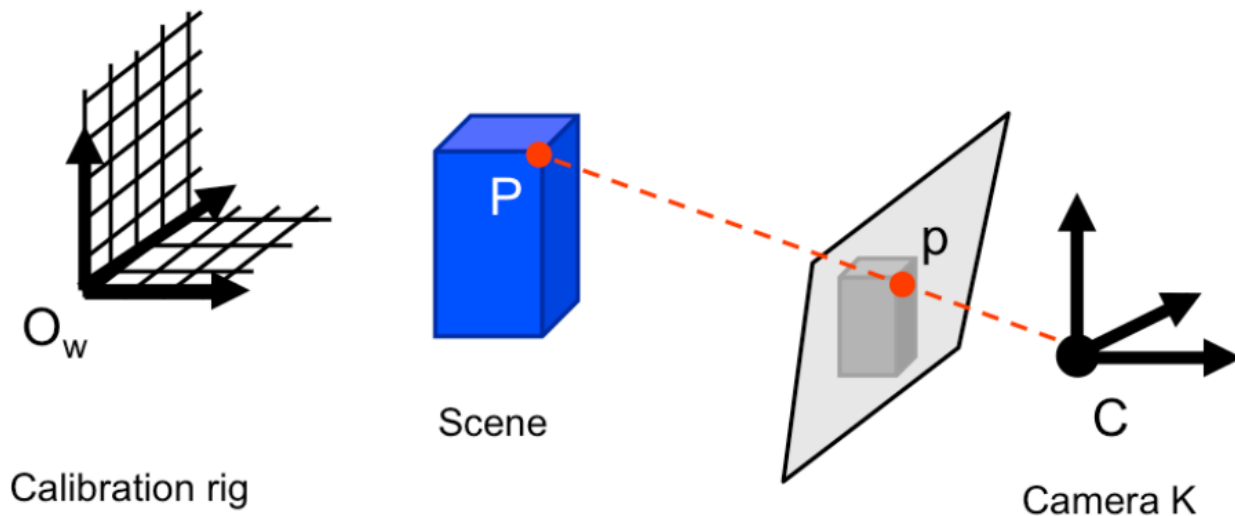
Epipolar Geometry



- Why stereo?
- Epipolar constraints
- Essential and Fundamental matrices
- Stereo Images rectification

- [FP] D. A. Forsyth and J. Ponce. Computer Vision: A Modern Approach (2nd Edition). Prentice Hall, 2011.
- [HZ] R. Hartley and A. Zisserman. Multiple View Geometry in Computer Vision. Cambridge University Press, 2003.
- **CS231A · Computer Vision: from 3D reconstruction to recognition, Prof. Silvio Savarese – Stanford University**

Without stereo...

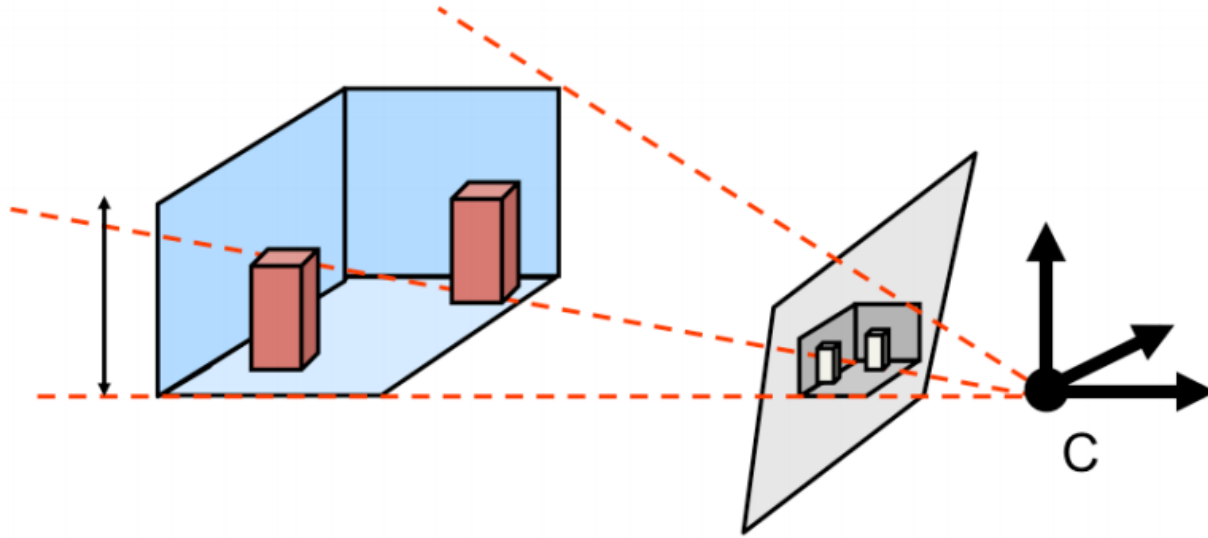


Calibration rig + rig position

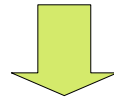


We can compute K and RT

Without stereo...



Vanishing lines and points + orthogonal planes

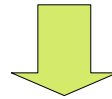


We can compute K

Without stereo...



Why it is **soooo** difficult?



Line of Sight: multiple potential correspondances for p in the 3D world..

Without stereo...

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Without stereo...

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Without stereo...

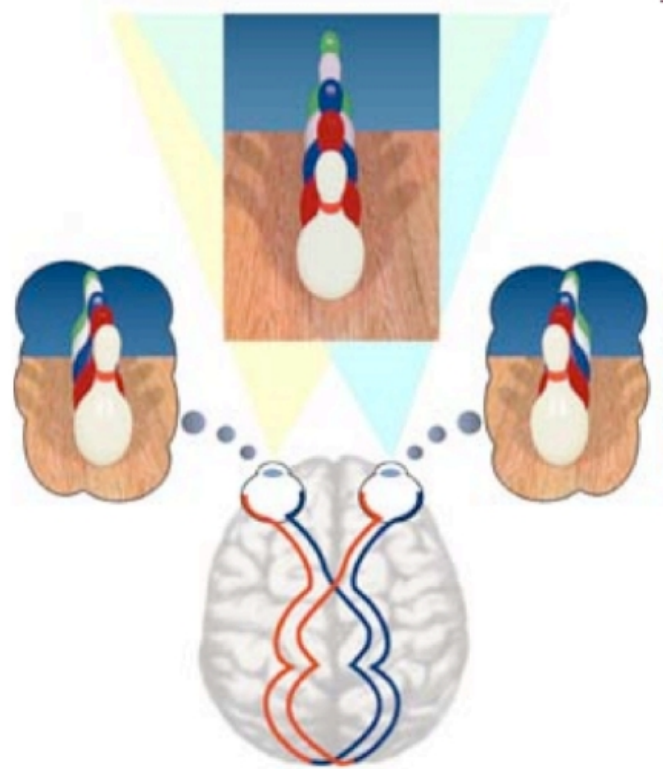


Without stereo...

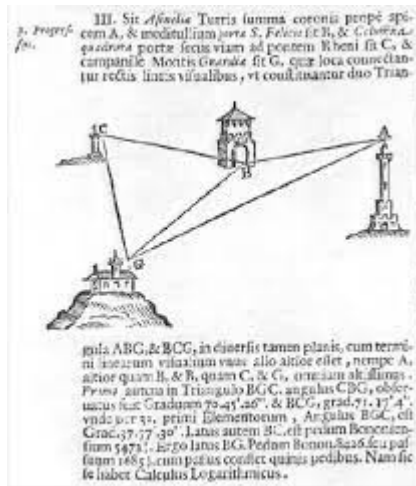
Nature and monocular vision



Stereo vision wins



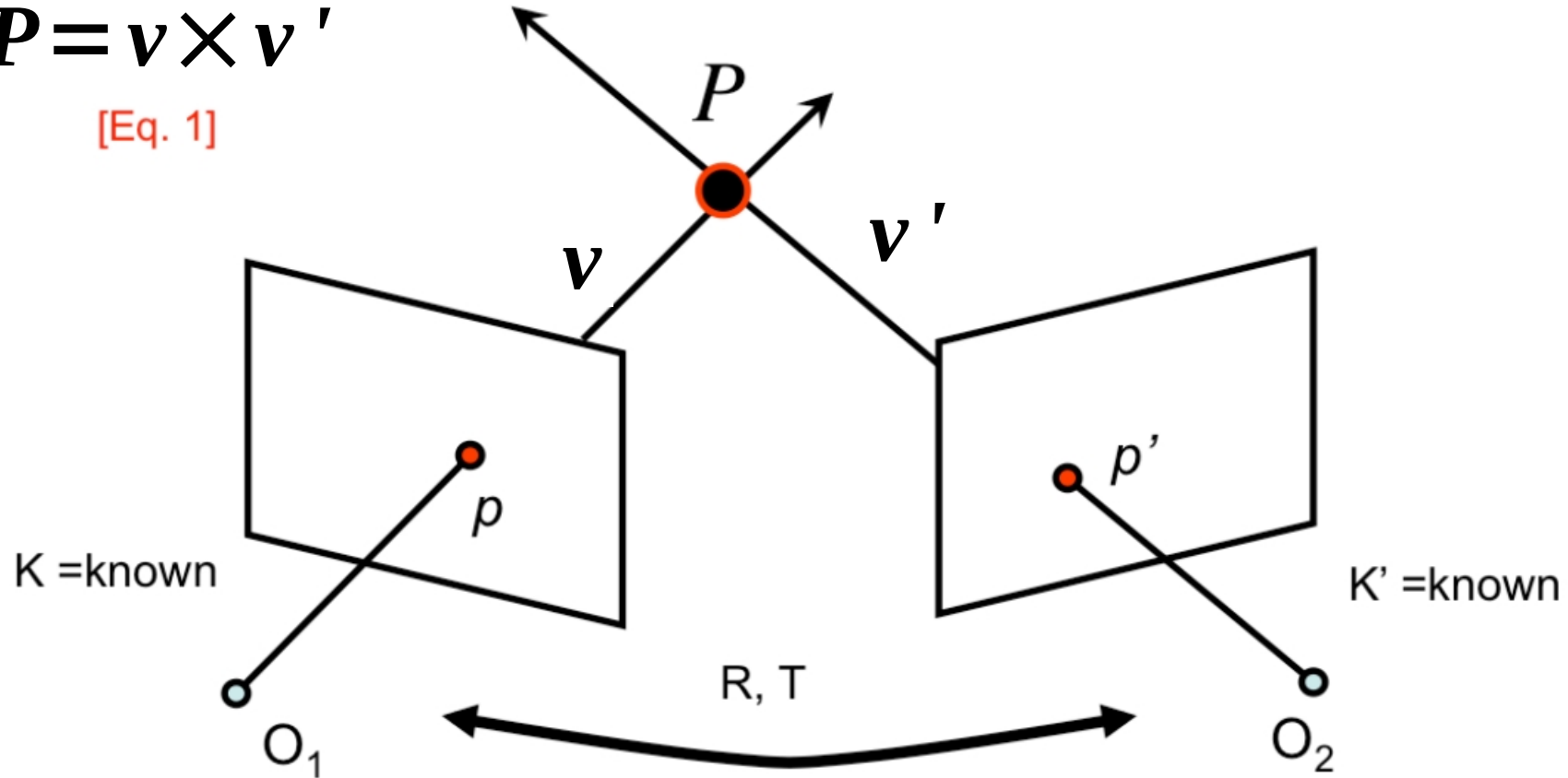
Triangulation



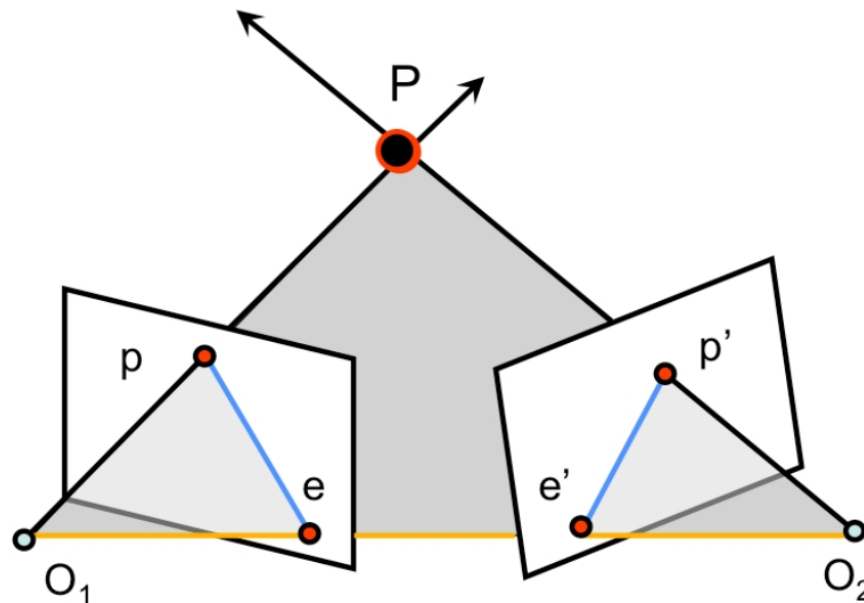
Triangulation

$$P = v \times v'$$

[Eq. 1]

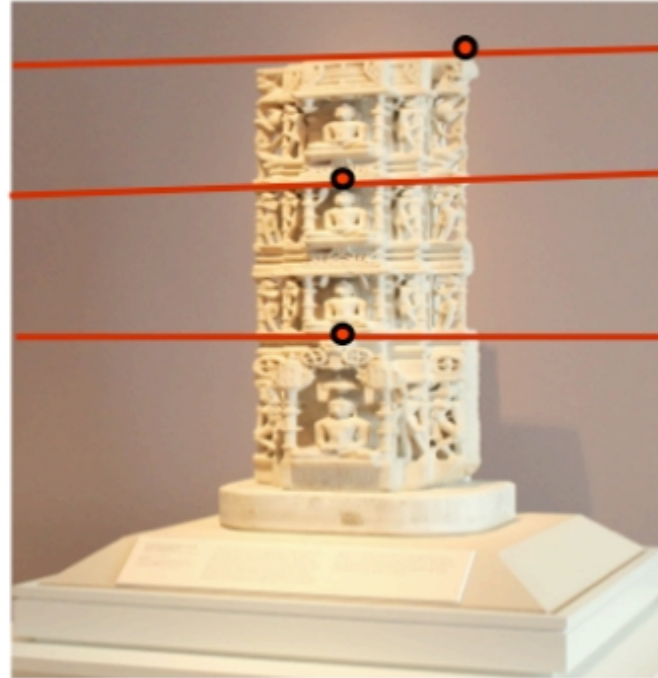
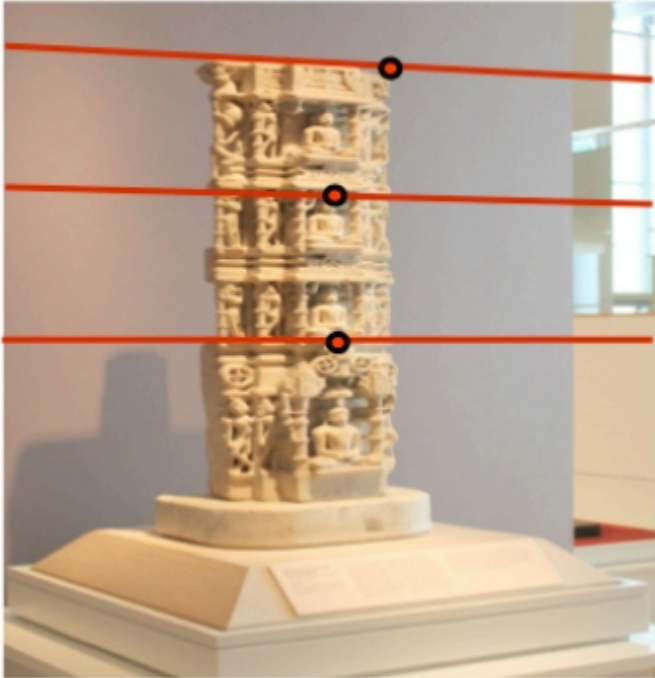


- **Correspondances:** given a p in one image find p' in the other image
- **Camera geometry:** given a set of correspondences find camera parameters
- **Scene Geometry:** given a set of correspondences and parameters of both cameras reconstruct the 3D scene



- O_1O_2P : epipolar plane
- O_1O_2 : baseline
- pe & $p'e'$: epipolar lines (all epipolar lines meet in e or e')
- e & e' : epipoles
 - Intersection of baseline with image planes
 - Projection of O_1 & O_2

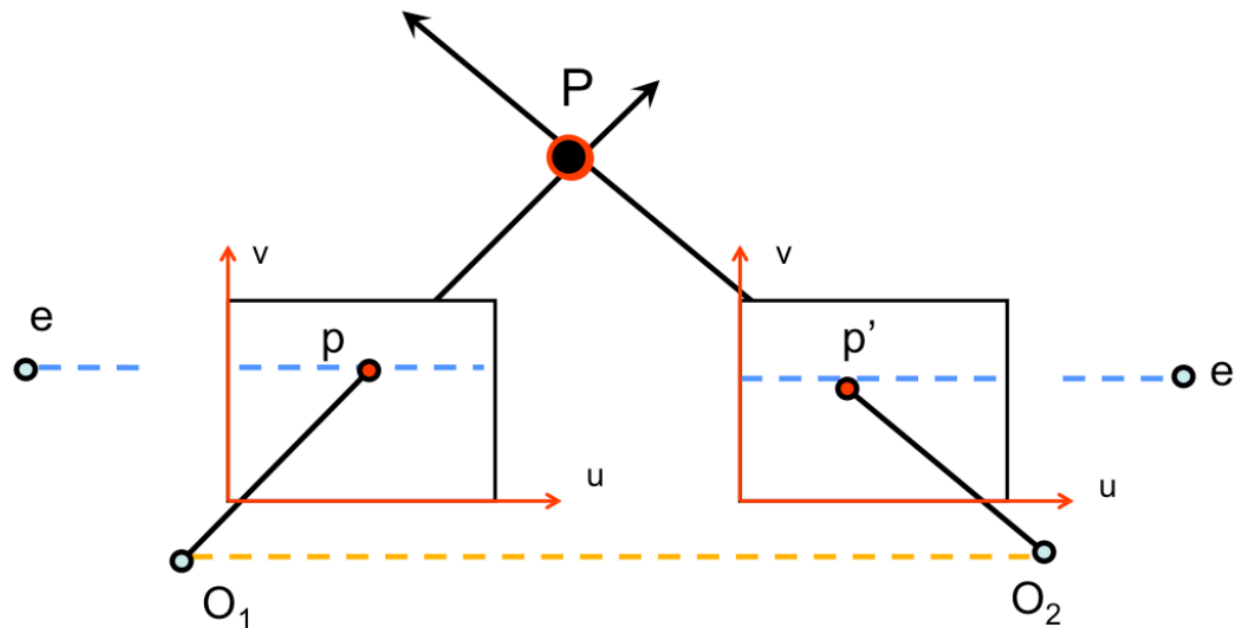
Epipolar lines



Epipolar lines



Special case: parallel image planes



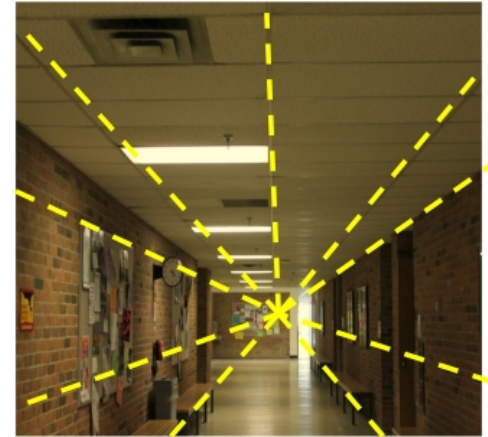
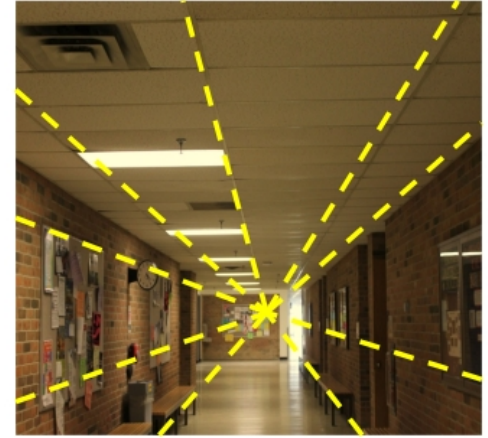
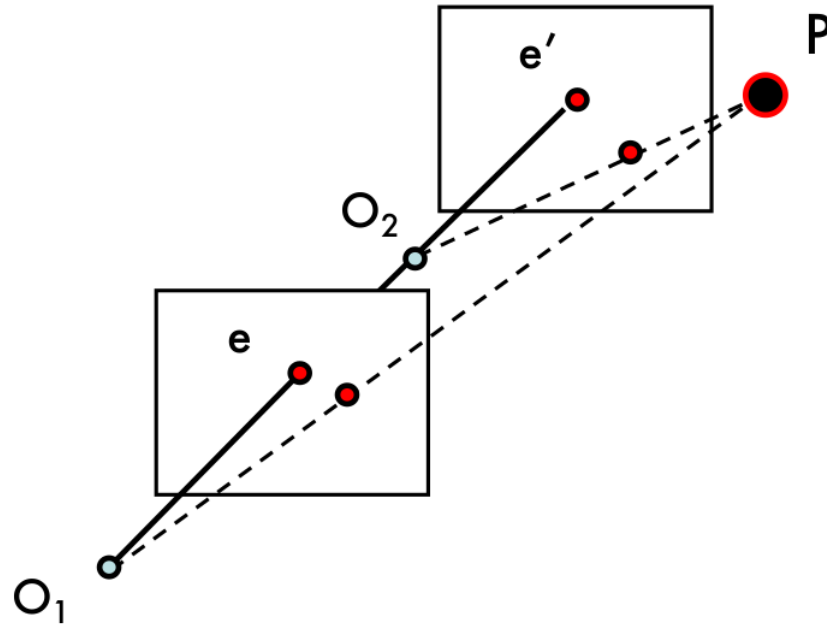
- Epipoles are at infinity
- Also intersection between baseline and image plane are at infinity
- Epipolar lines are parallel to u axis

Special case: parallel image planes

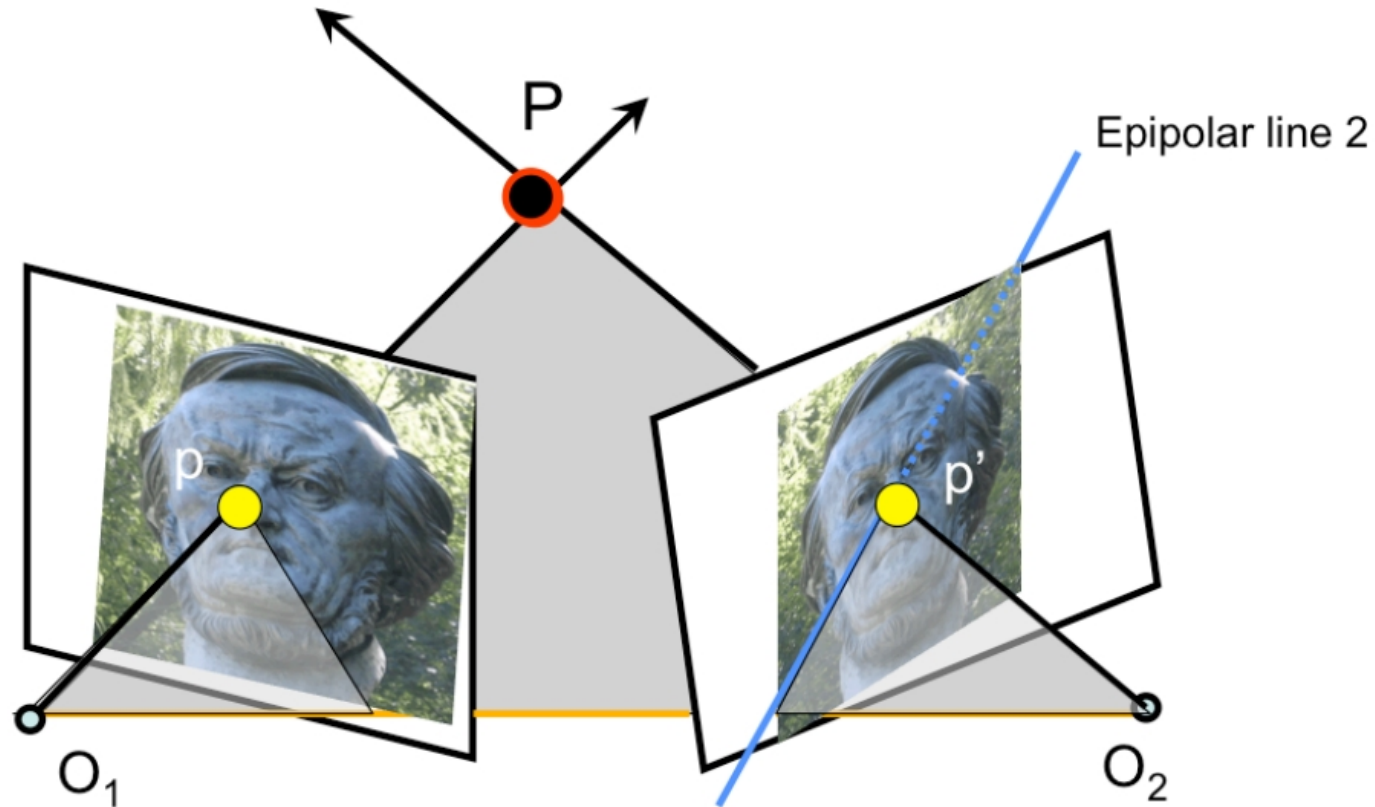


- Example of two images whose planes are parallel to each other
- Epipolar lines are parallel to each other

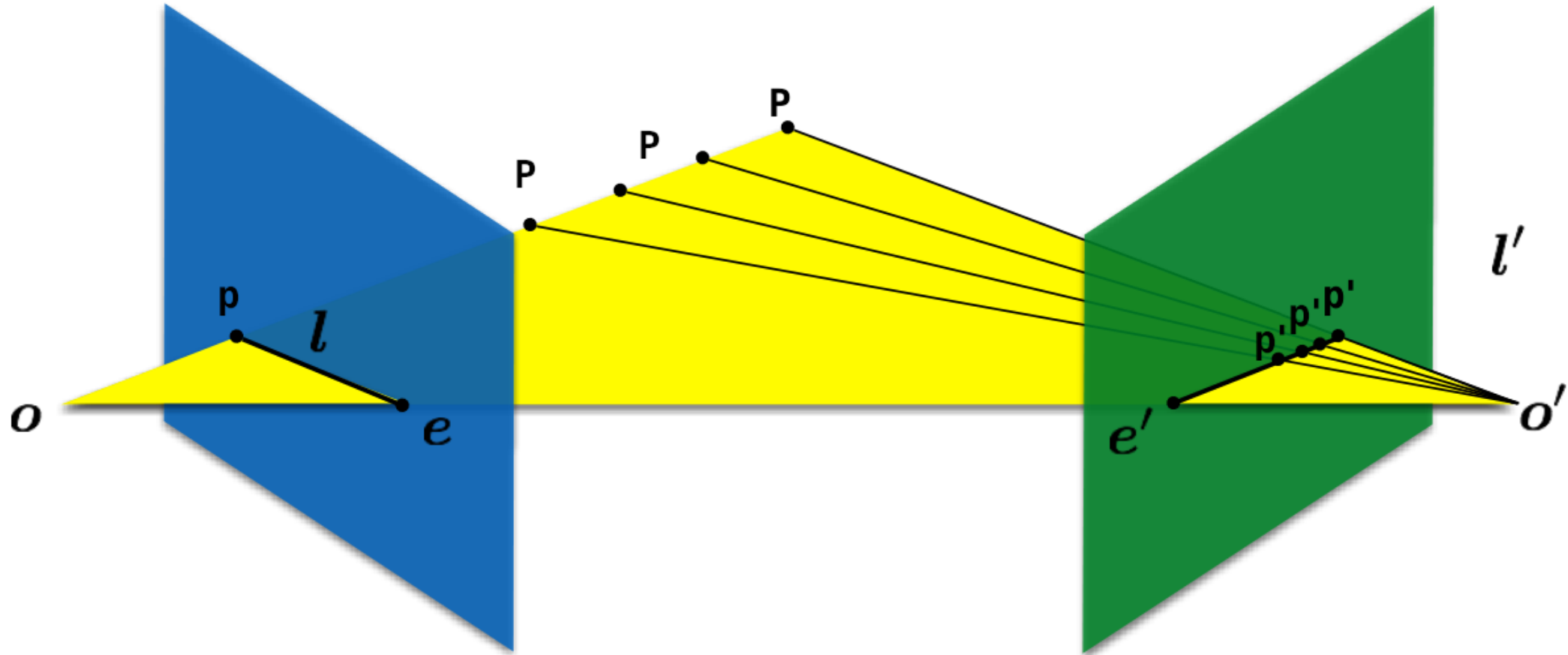
Special case: forward translation



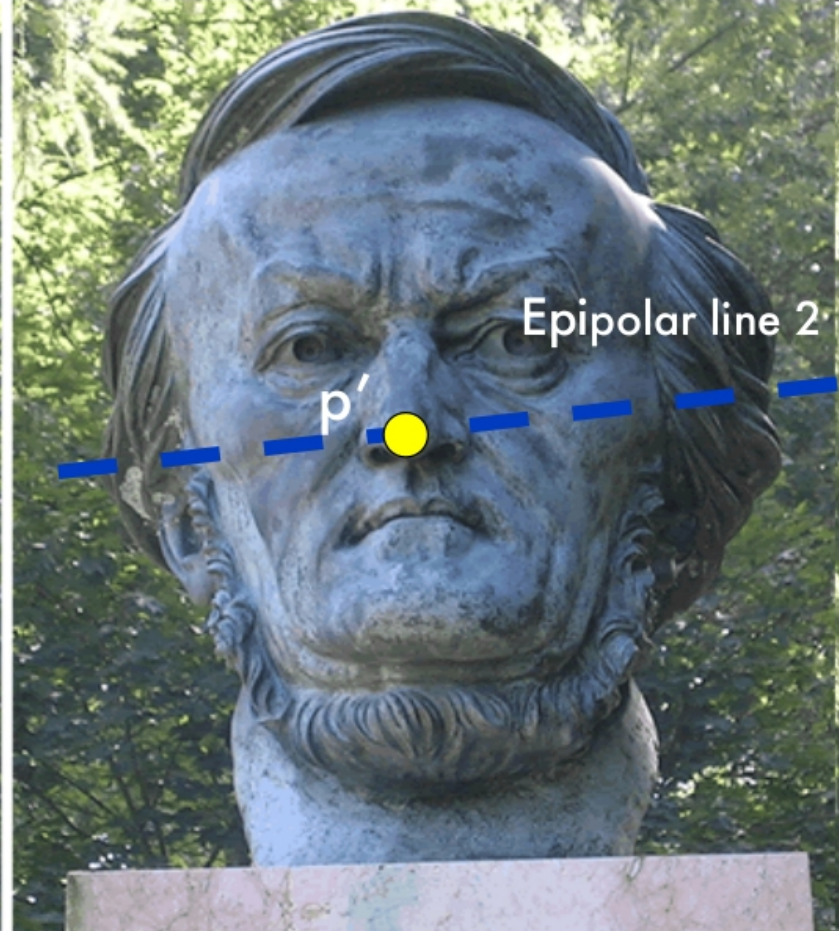
Epipolar constraint



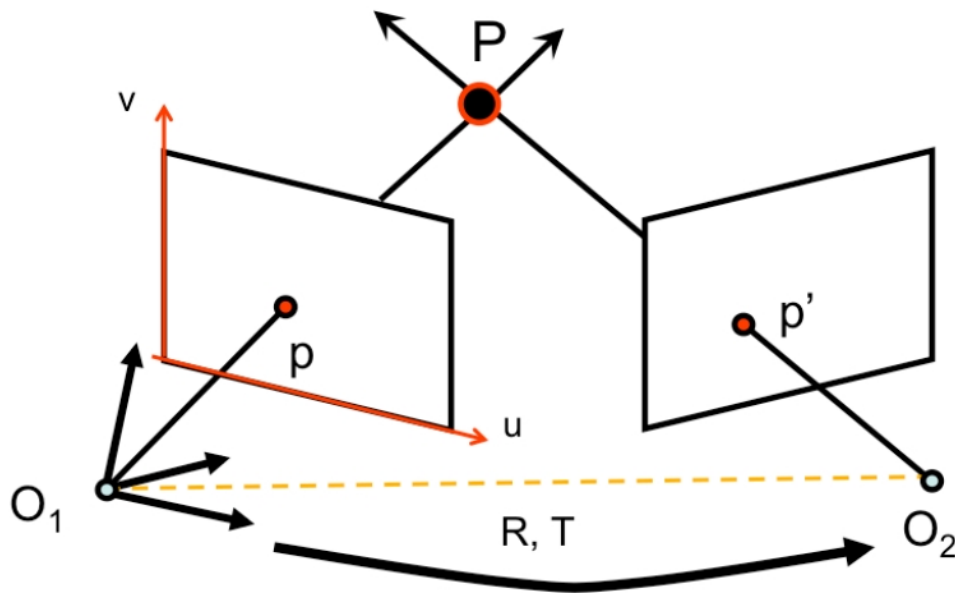
Epipolar constraint



Epipolar constraint



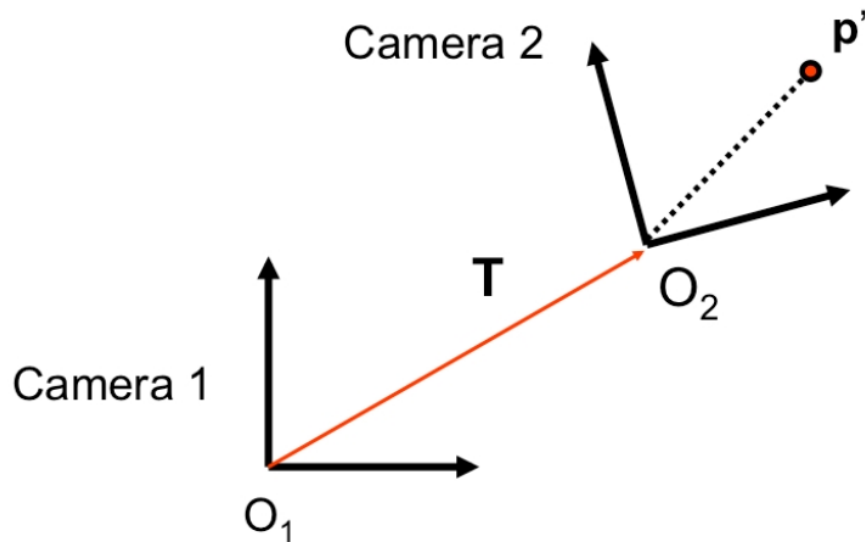
Epipolar constraint



$$M = K \begin{bmatrix} I & 0 \end{bmatrix}$$
$$M P = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = p \quad \text{[Eq. 3]}$$

$$M' = K' \begin{bmatrix} R^T & -R^T T \end{bmatrix}$$
$$M' P = \begin{bmatrix} u' \\ v' \\ 1 \end{bmatrix} = p' \quad \text{[Eq. 4]}$$

Relative position for cameras



$$p'_{o_1} = R p'_{o_2} + T$$

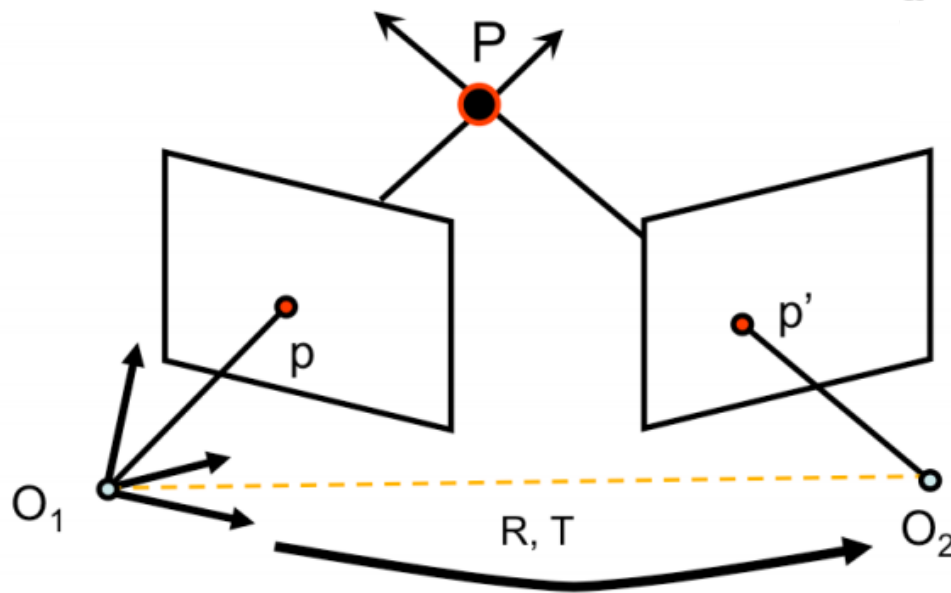
$$p'_{o_2} = R^T (p'_{o_1} - T)$$

- T is O_2 in the O_1 reference system
 - Actual translation
- R is the rotation matrix such as **free vector** p' for O_2 is Rp' for O_1
 - Rotation limited to epipolar plane

- Is a camera where $K = I$
- Not a real camera but a simplified model that can ease to understand epipolar relations
- Actually used when we know the K and K' of real cameras
 - i.e. the cameras are already calibrated

$$K_{canonical} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad p = Kp_c$$

Epipolar constraint



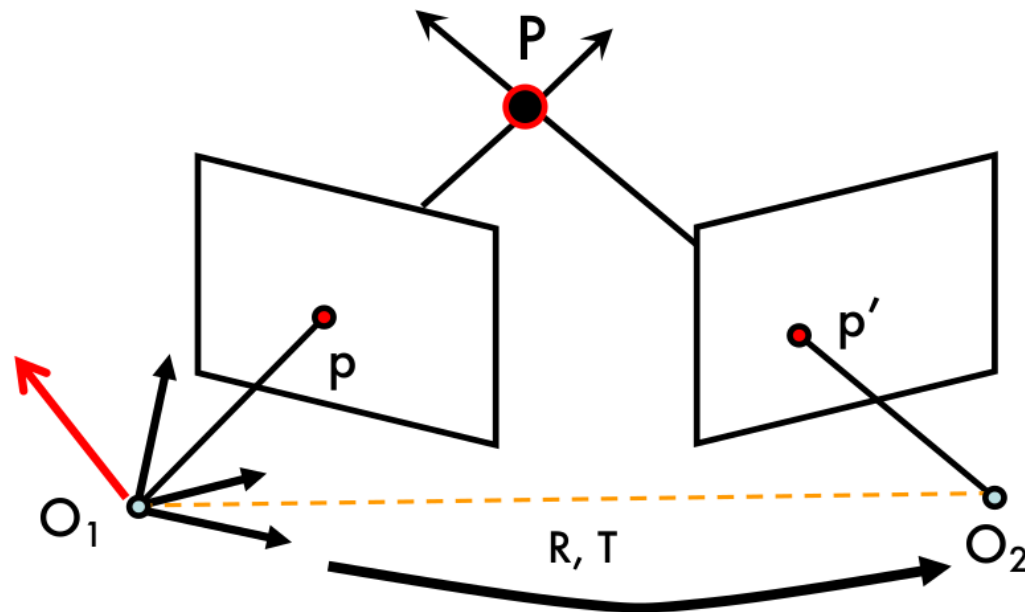
$$K_{canonical} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = K \begin{bmatrix} I & 0 \end{bmatrix} \quad \boxed{K = K' \text{ are known (canonical cameras)}} \quad M' = K' \begin{bmatrix} R^T & -R^T T \end{bmatrix}$$

$\downarrow$
 $\downarrow$

$$M = \begin{bmatrix} I & 0 \end{bmatrix} \quad [\text{Eq. 5}] \qquad M' = \begin{bmatrix} R^T & -R^T T \end{bmatrix} \quad [\text{Eq. 6}]$$

Epipolar constraint

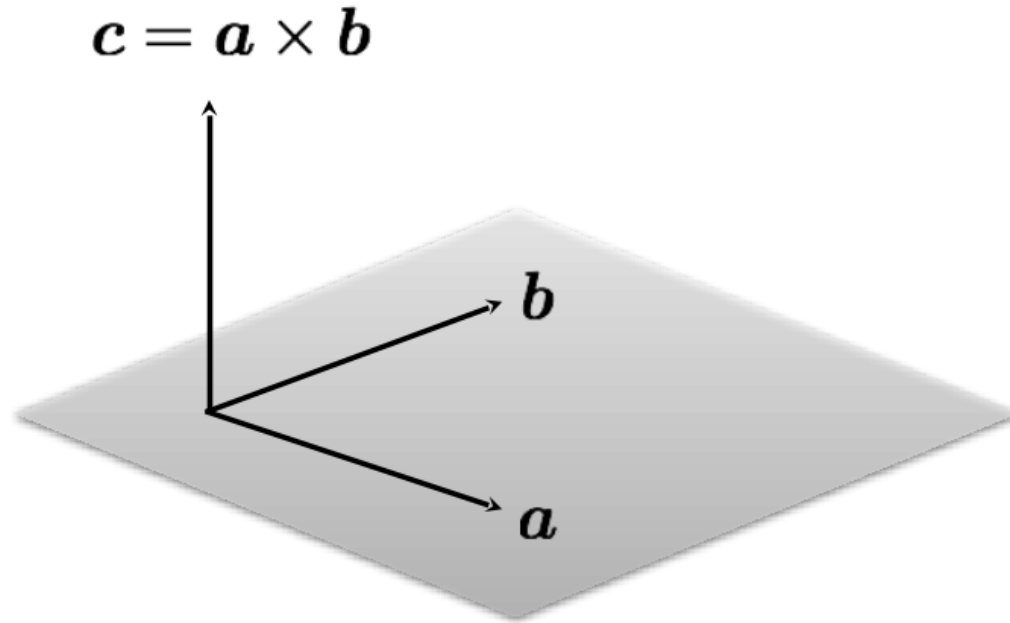


p' in first camera reference system is $= R p' + T$

$T \times ((R p') + T) = T \times (R p')$ is perpendicular to epipolar plane

$$\rightarrow p^T \cdot [T \times (R p')] = 0 \quad \text{[Eq. 7]}$$

Cross product recall



$$c \cdot a = 0$$

$$c \cdot b = 0$$

$$a \times b = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}$$

cross product of two vectors
in the same direction is zero

$$a \times a = 0$$

remember this!!!

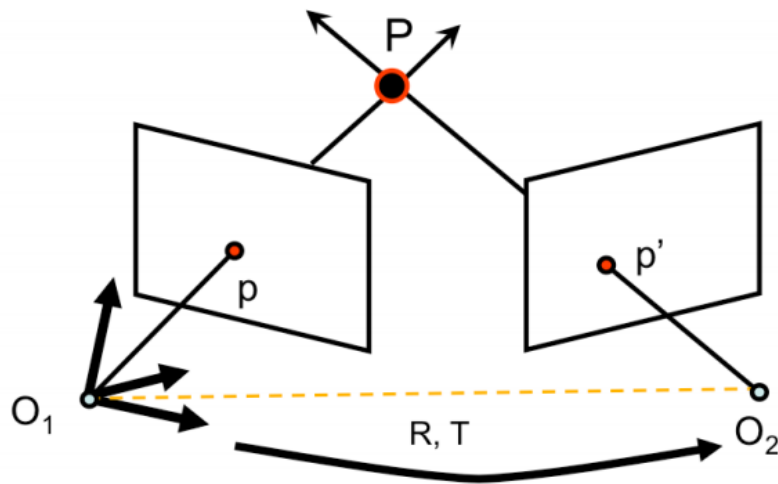
Credits: Ioannis Gkioulekas

Cross product as matrix multiplication

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & -a_x & 0 \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = [\mathbf{a}_x] \mathbf{b}$$

$$\mathbf{a} \times \mathbf{b} = [\mathbf{a}_x] \mathbf{b}$$

$$\mathbf{T} \times (\mathbf{R} \mathbf{p} ') = [\mathbf{T}_x] (\mathbf{R} \mathbf{p} ') = ([\mathbf{T}_x] \mathbf{R}) \mathbf{p} '$$

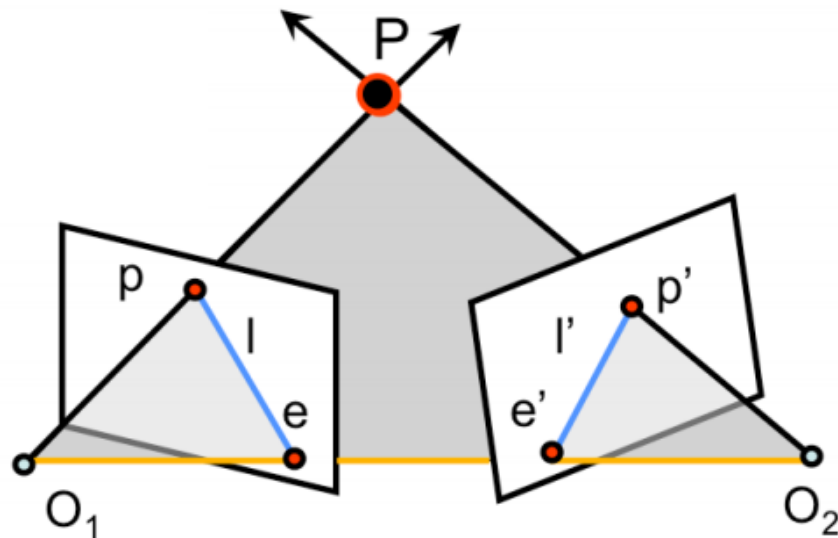


$$p^T \cdot [T \times (Rp')] = 0$$

$$p^T \cdot \underbrace{\left(\begin{bmatrix} T_x \end{bmatrix} R \right)}_E p' = 0$$

- E is the **Essential Matrix**

Essential Matrix: features



$$p^T \cdot E p' = 0$$

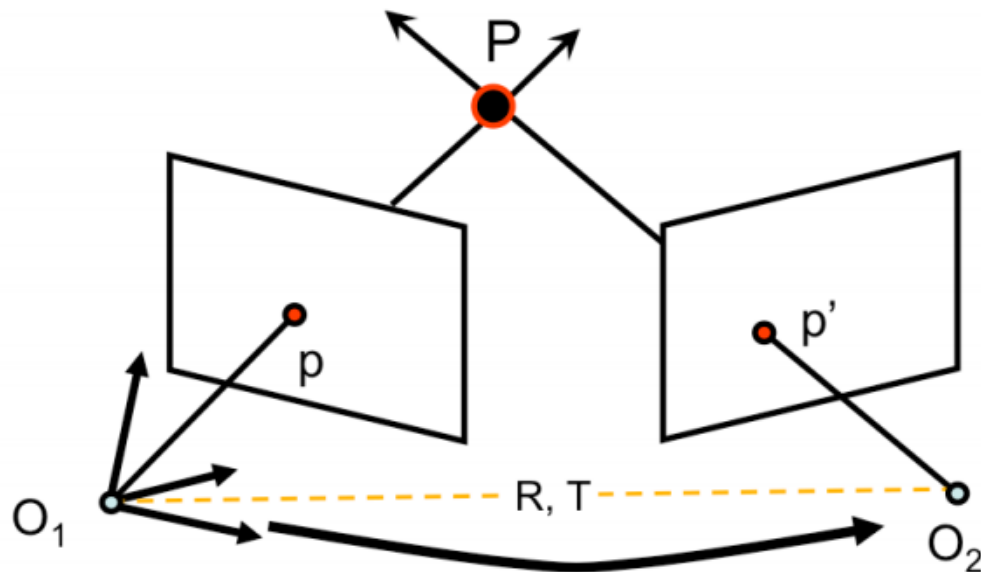
- $l = E p' \rightarrow$ epipolar line for p
- $l' = E^T p \rightarrow$ epipolar line for p'
- $E e' = 0 \quad E^T e = 0 \rightarrow$ namely epipoles are solution for homogeneous equation
- E is a 3×3 matrix with 5 DOF
- E is singular and its rank is 2

$$\mathbf{p}^T \mathbf{E} \mathbf{p}' = 0 \quad \text{Longuet-Higgins Equation}$$

- Usually we do not use ideal cameras
- Is then $\mathbf{E}$ so useful?
- Yes if we think about normalized camera coordinates and not image coordinates
 - For image coordinates we need to know $\mathbf{K}$

$$\mathbf{p} = \mathbf{K} \mathbf{p}_c \quad \mathbf{p}_c = \mathbf{K}^{-1} \mathbf{p}$$

Epipolar constraint (non ideal camera)



$$M = K \begin{bmatrix} I & 0 \end{bmatrix}$$

$$p_c = K^{-1} p \quad [\text{Eq. 11}]$$

$$M' = K' \begin{bmatrix} R^T & -R^T T \end{bmatrix}$$

$$p'_c = K'^{-1} p' \quad [\text{Eq. 12}]$$

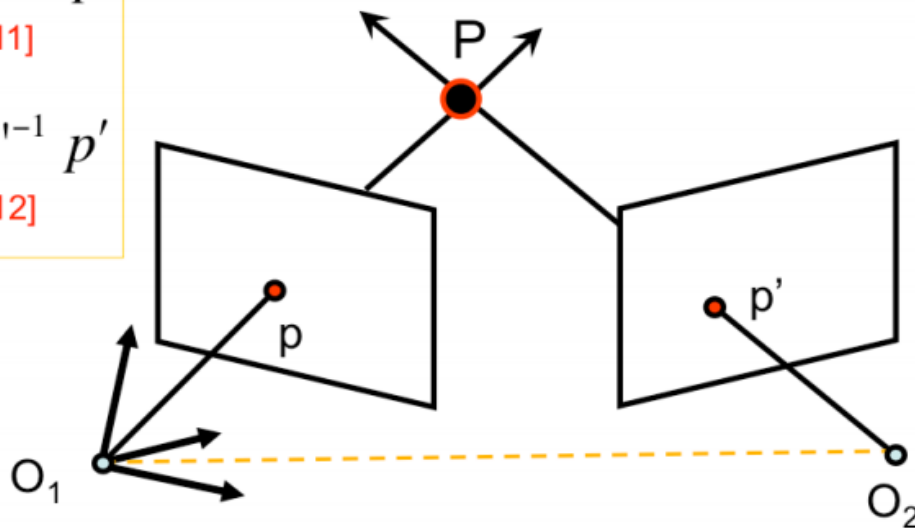
Epipolar constraint (non ideal camera)

$$p_c = K^{-1} p$$

[Eq. 11]

$$p'_c = K'^{-1} p'$$

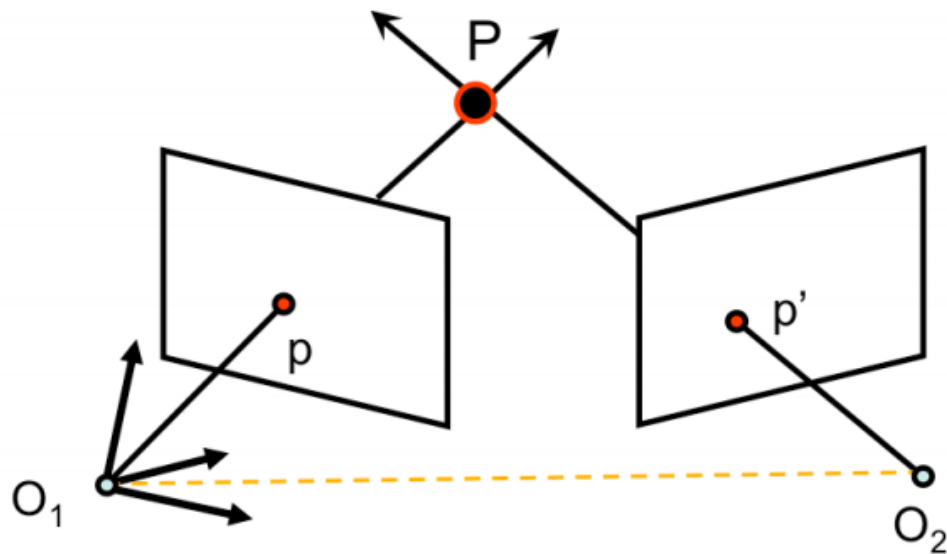
[Eq. 12]



[Eq.9]

$$p_c^T \cdot [T_x] \cdot R p'_c = 0 \rightarrow (K^{-1} p)^T \cdot [T_x] \cdot R K'^{-1} p' = 0$$

$$p^T \boxed{K^{-T} \cdot [T_x] \cdot R K'^{-1}} p' = 0 \rightarrow p^T \boxed{F} p' = 0 \quad \text{[Eq. 13]}$$



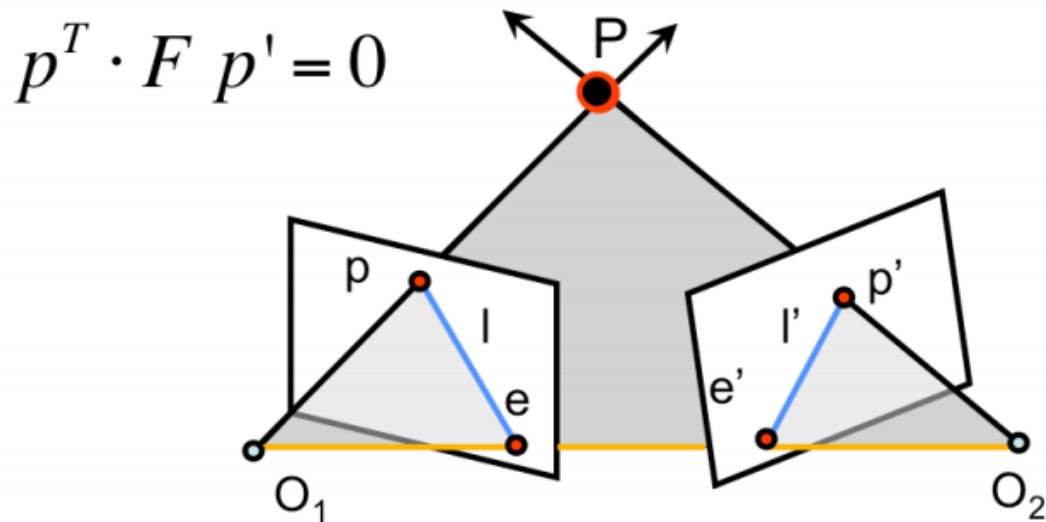
[Eq. 13]

$$p^T F p' = 0$$

$$F = K^{-T} \cdot [T_x] \cdot R K'^{-1}$$

[Eq. 14]

- $F \rightarrow$ Fundamental Matrix (Faugeras and Luong 1992)



- $l = Fp' \rightarrow$ epipolar line for p
- $l' = F^T p \rightarrow$ epipolar line for p'
- $Fe' = 0 \quad F^T e = 0 \rightarrow$ namely epipoles are solution for homogeneous equation
- F is a 3×3 matrix with 7 DOF
- F is singular and its rank is 2

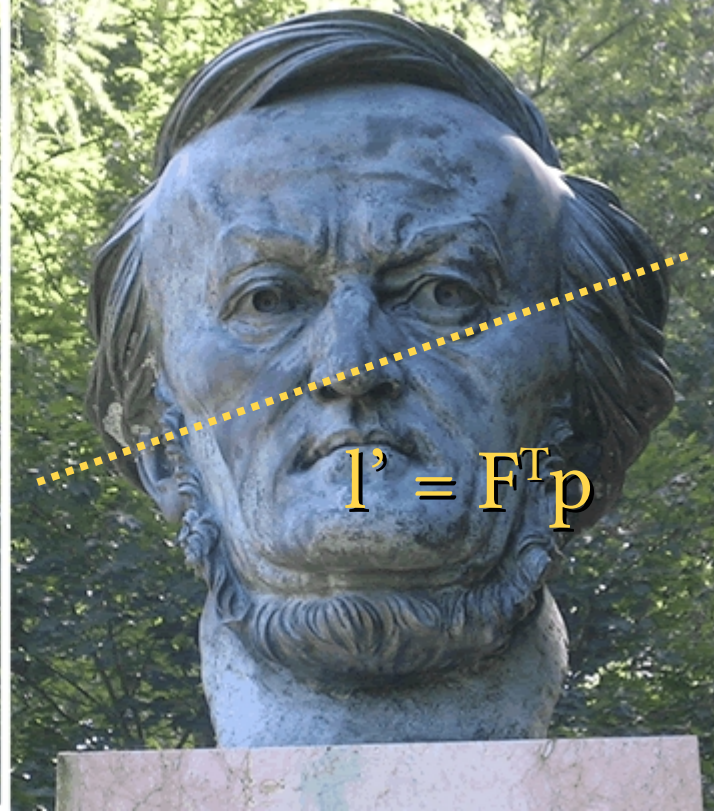
$$p_c^T E p'_c = 0 \quad p^T F p' = 0$$

$$p_c = K^{-1} p \quad p'_c = K'^{-1} p'$$

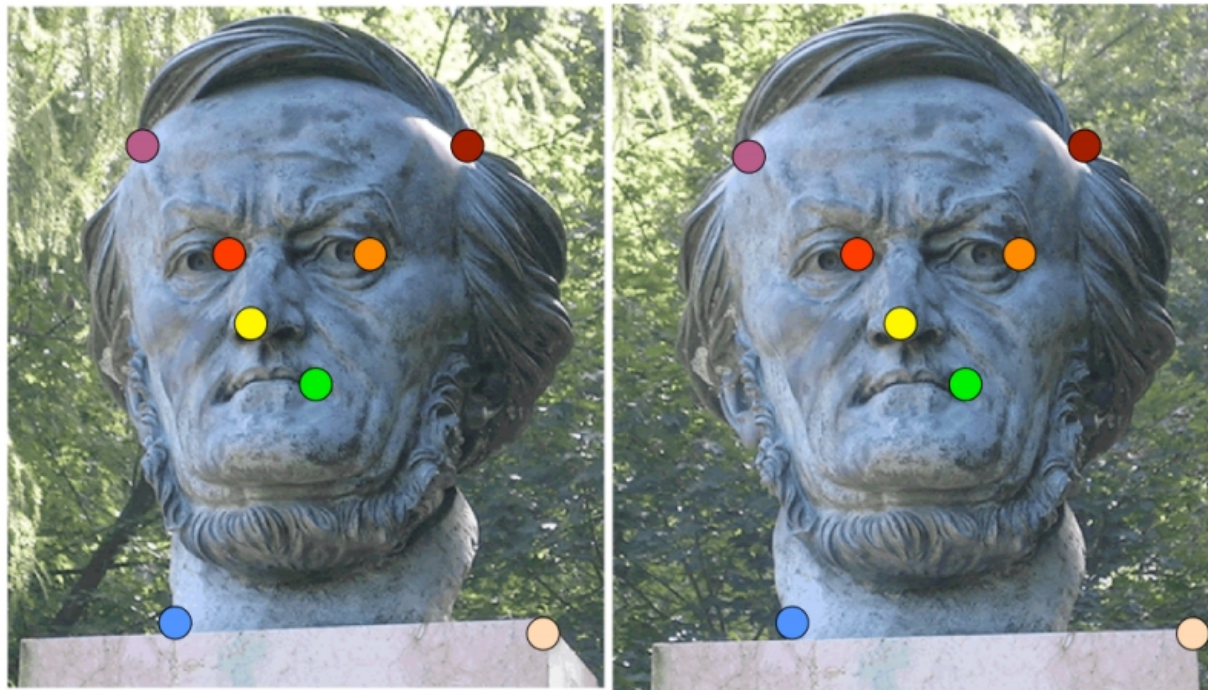
$$p^T K^{-T} E K^{-1} p' = 0$$

$$K^{-T} E K^{-1} = F$$

Why we want F (or E)?



Fundamental Matrix estimation



- Can we estimate F without knowing nothing about K , R , T ?
- Yes $\rightarrow$ 8-point algorithm

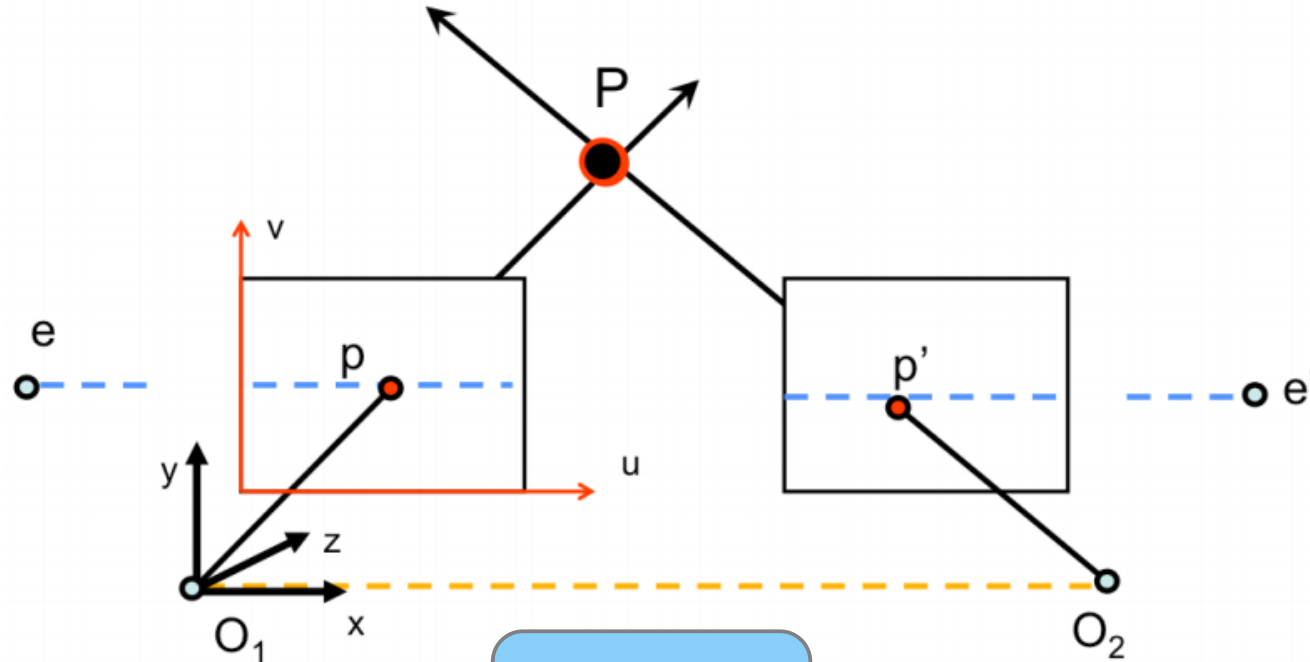
- Why those matrices are important?
- They give us a relation between two stereo pairs even without knowing anything about their calibration
 - The 8 points approach can be used to estimate them
- In the next lesson we are going to see how this, limited, knowledge anyway allows us to reconstruct a 3D scene



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Special Case Parallel Image Planes

Special case: parallel image planes



$$K_1 = K_2 = \text{known}$$

x parallel to $O_1 O_2$

$$\mathbf{E} = ??$$

$$R = I$$

$$T = (T_x, 0, 0)$$

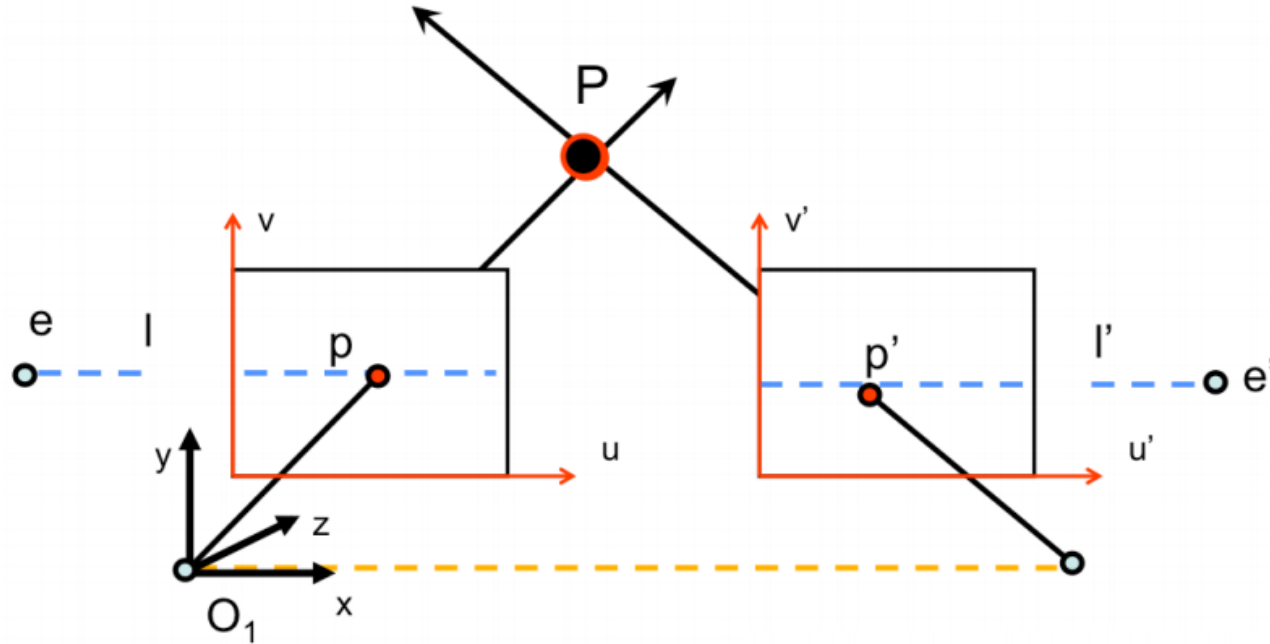
Special case: parallel image planes

$$E = [T_x] \cdot R$$

$$R = I \quad T = \begin{bmatrix} T_x & 0 & 0 \end{bmatrix} = \begin{bmatrix} T & 0 & 0 \end{bmatrix}$$

$$E = \begin{bmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & T_x & 0 \end{bmatrix} \cdot I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -T \\ 0 & T & 0 \end{bmatrix}$$

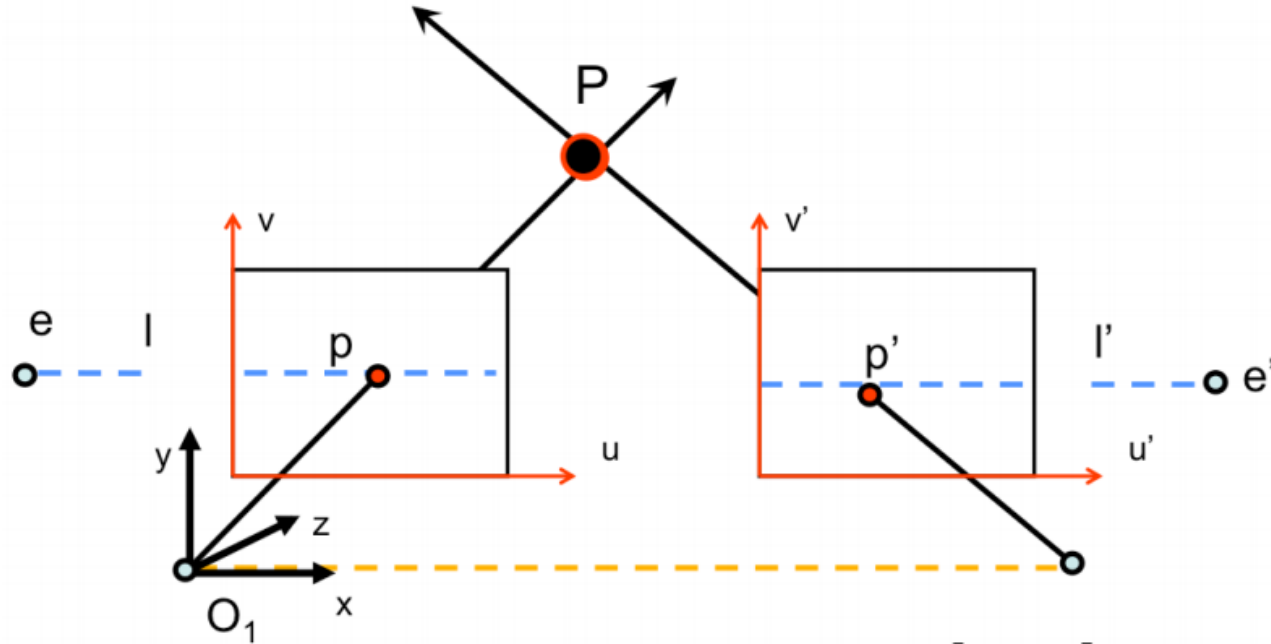
Special case: parallel image planes



- Epipolar lines are horizontal

$$l = E p' = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -T \\ 0 & T & 0 \end{bmatrix} \begin{bmatrix} u' \\ v' \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -T \\ T v' \end{bmatrix} \text{ horizontal!}$$

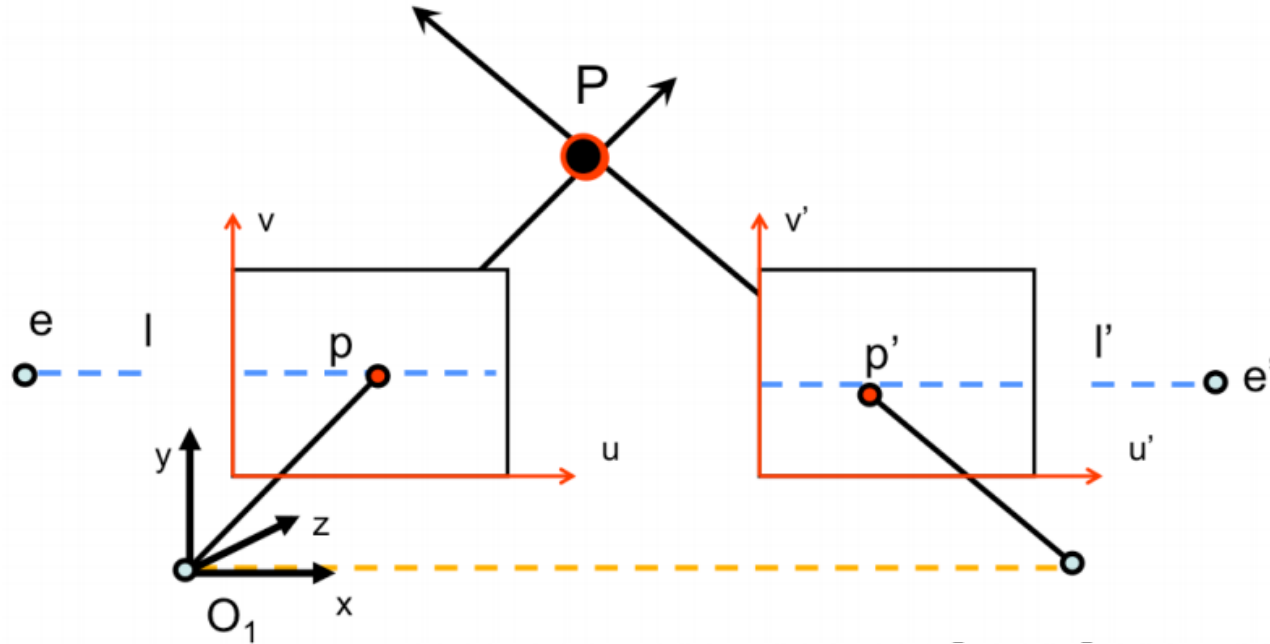
Special case: parallel image planes



- How p & p' are related?

$$p^T \cdot E p' = 0$$

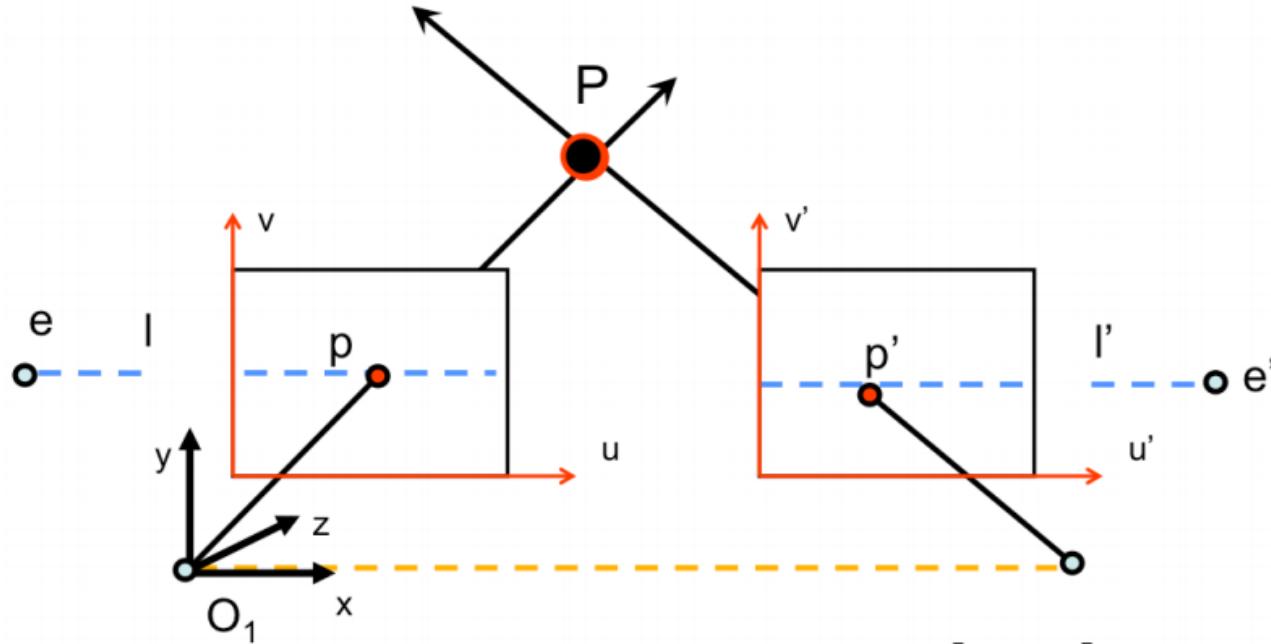
Special case: parallel image planes



- How p & p' are related?

$$(u \ v \ 1) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -T \\ 0 & T & 0 \end{bmatrix} \begin{pmatrix} u' \\ v' \\ 1 \end{pmatrix} = 0 \Rightarrow (u \ v \ 1) \begin{pmatrix} 0 \\ -T \\ Tv' \end{pmatrix} = 0 \Rightarrow Tv = Tv' \Rightarrow v = v'$$

Special case: parallel image planes



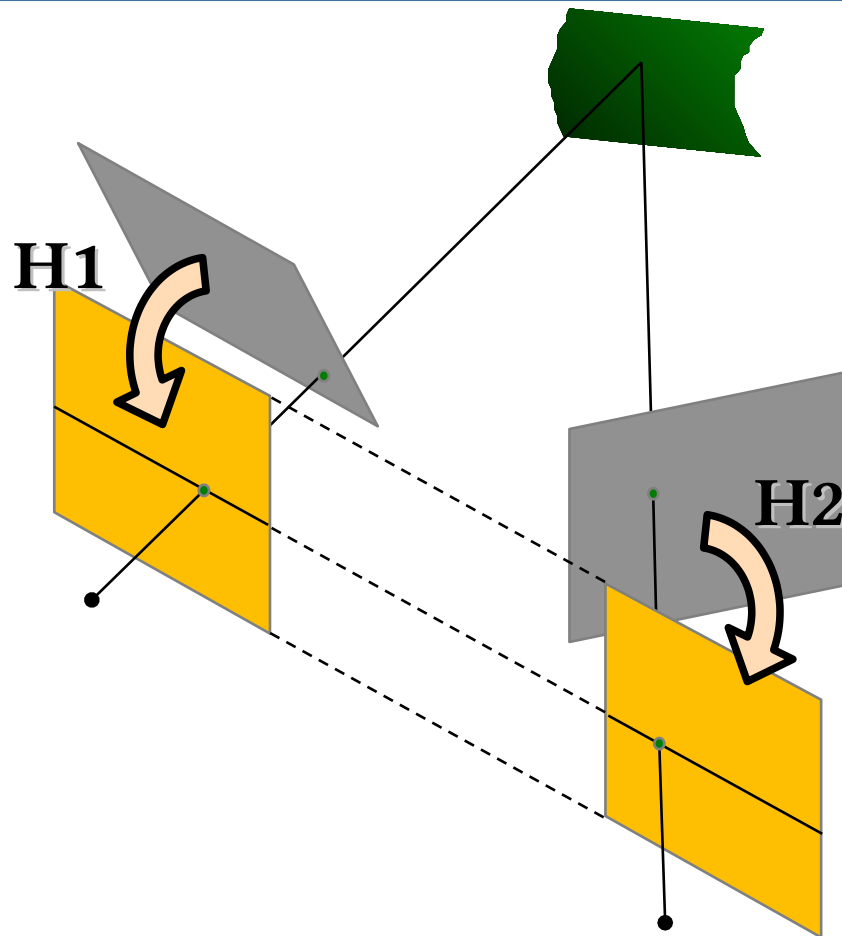
- $v=v'$ means the same line!
- The search for p' is simpler...

True for canonical cameras!

Also when $K=K'$!

Image Rectification

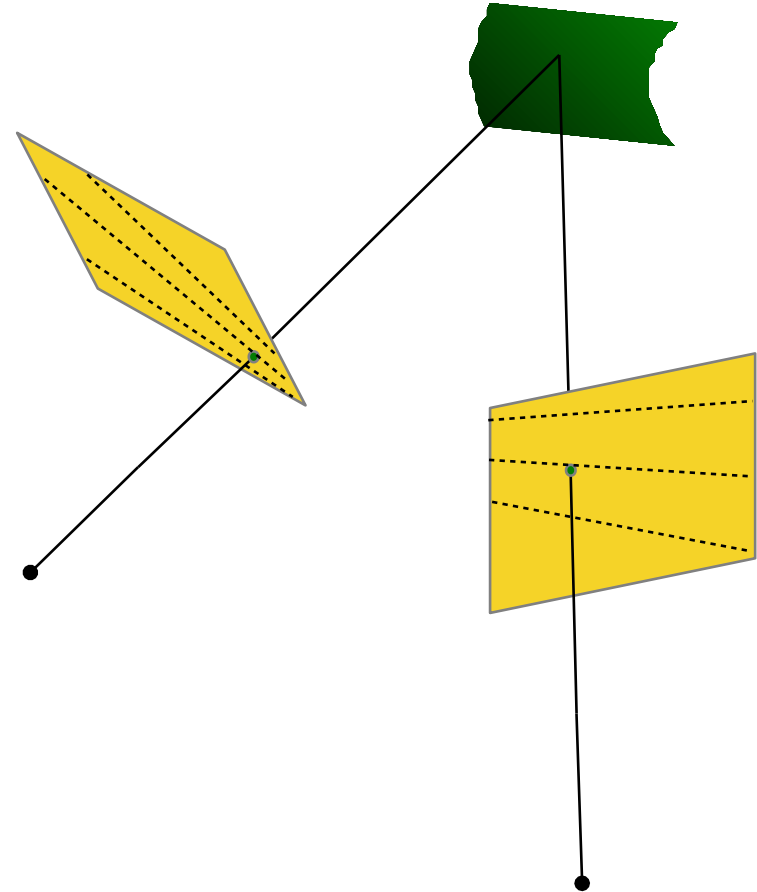
- Parallel image planes are better to handle for stereovision
- Simply accurately align cameras
 - Good idea, but it does not work!
 - Precision, noise, thermal effects...
- Reproject camera planes onto a common plane parallel to the line between camera centers



- Assuming $K=K'$ as known
 - 1) Compute $E \rightarrow$ we get R
 - 2) Rotate right image by R
 - 3) Rotate both images to move e and e' to ∞
 - We need to compute a R_{rect}
 - 4) Adjuste the scale

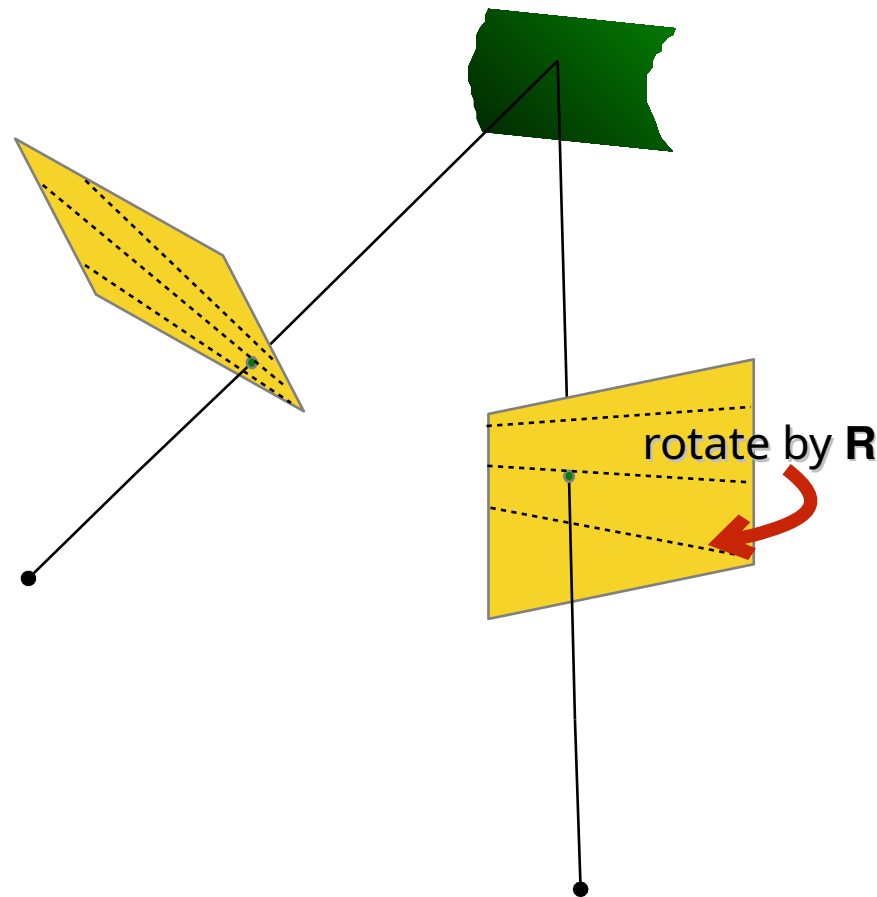
Stereo Image Rectification

1) Compute $E \rightarrow$ we get R



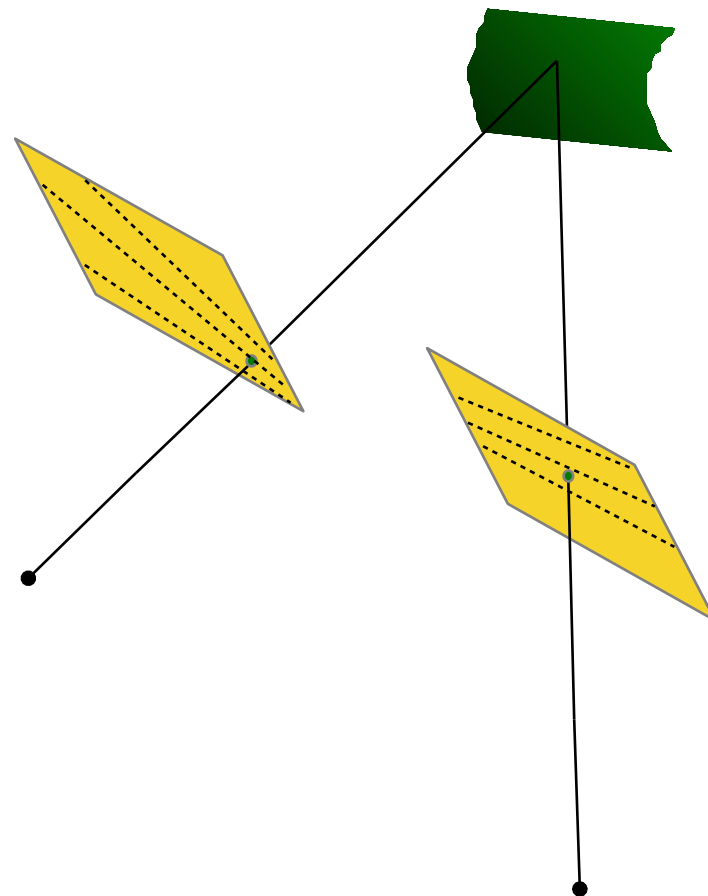
Stereo Image Rectification

- 1) Compute $E \rightarrow$ we get R
- 2) Rotate left image by R



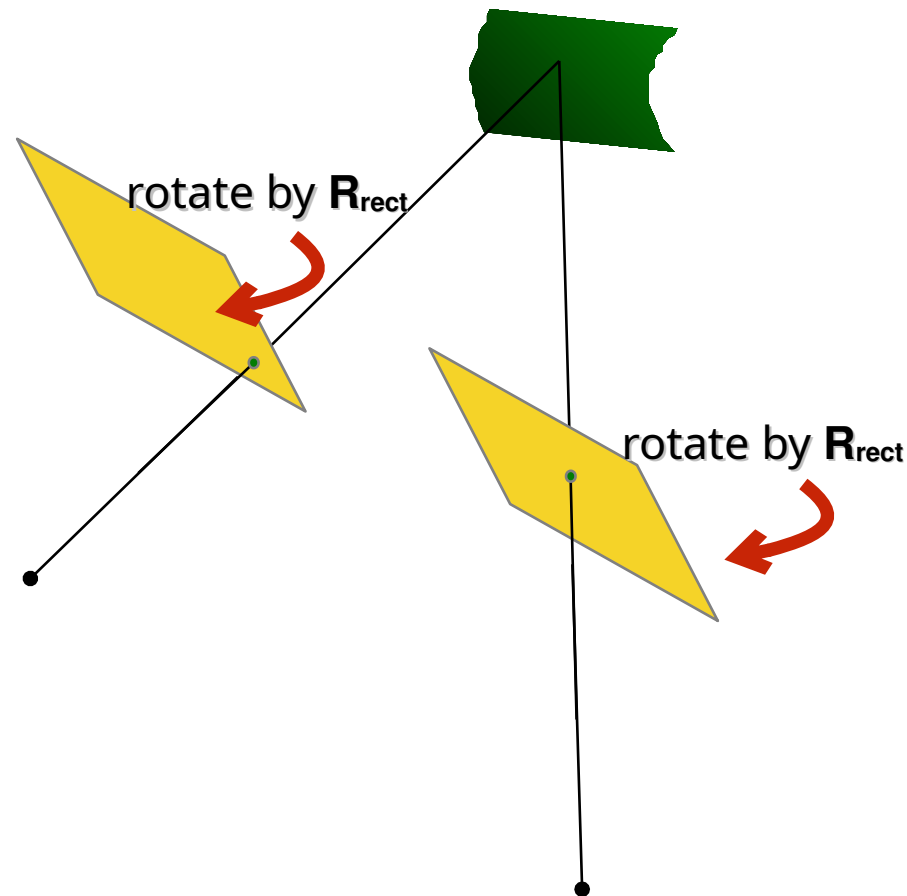
Stereo Image Rectification

- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R



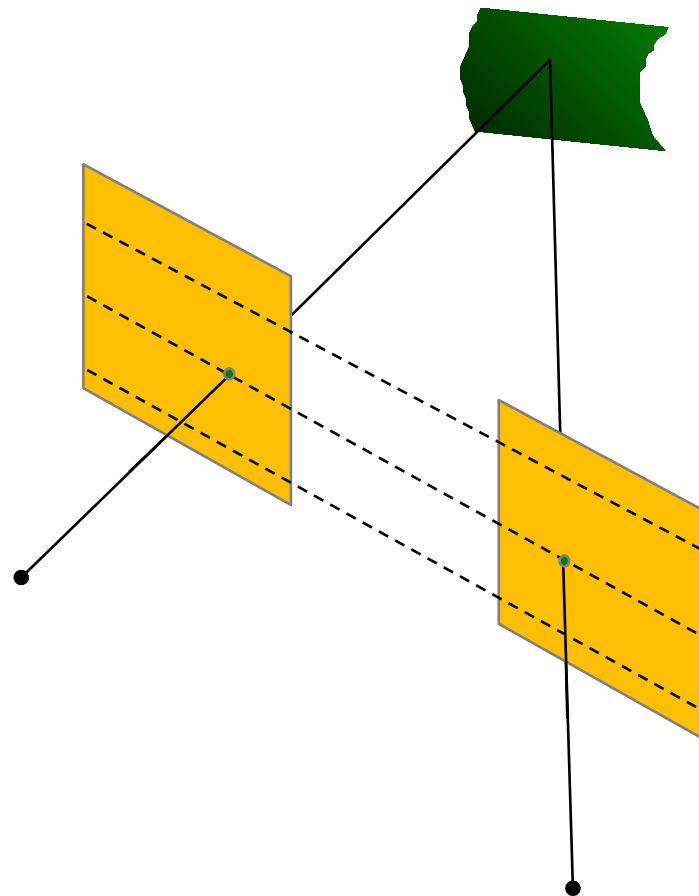
Stereo Image Rectification

- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R
- 3) Rotate both images by R_{rect}



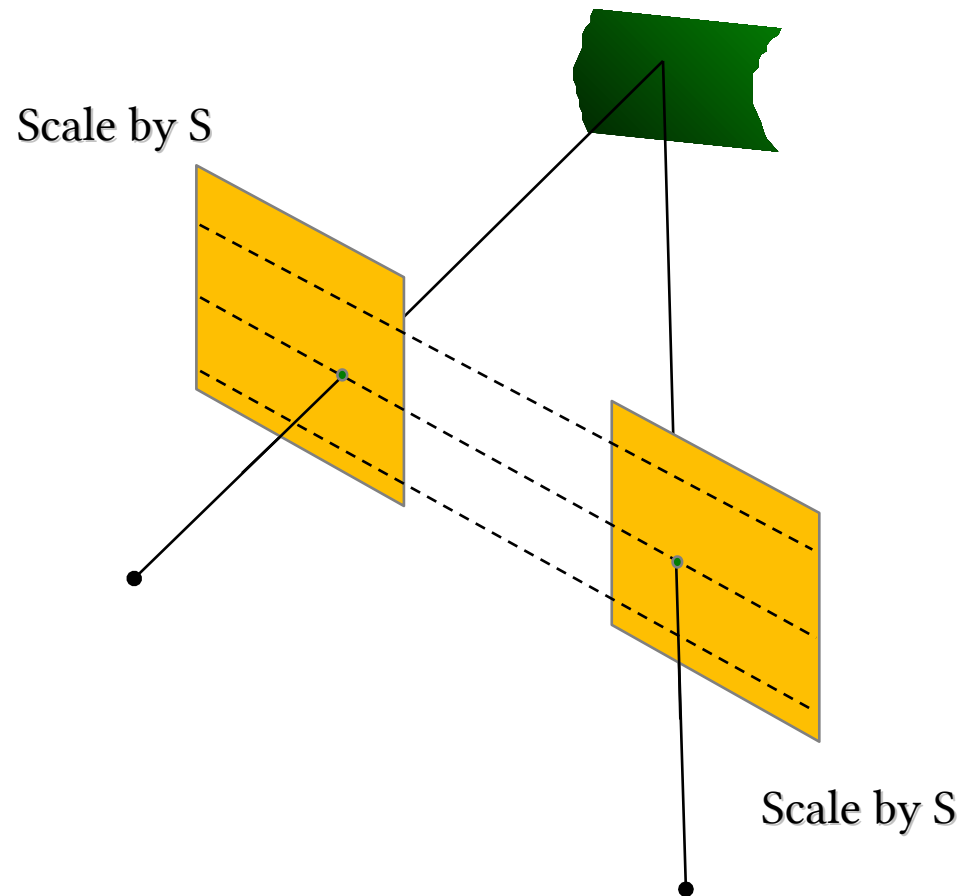
Stereo Image Rectification

- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R
- 3) Rotate both images by R_{rect}



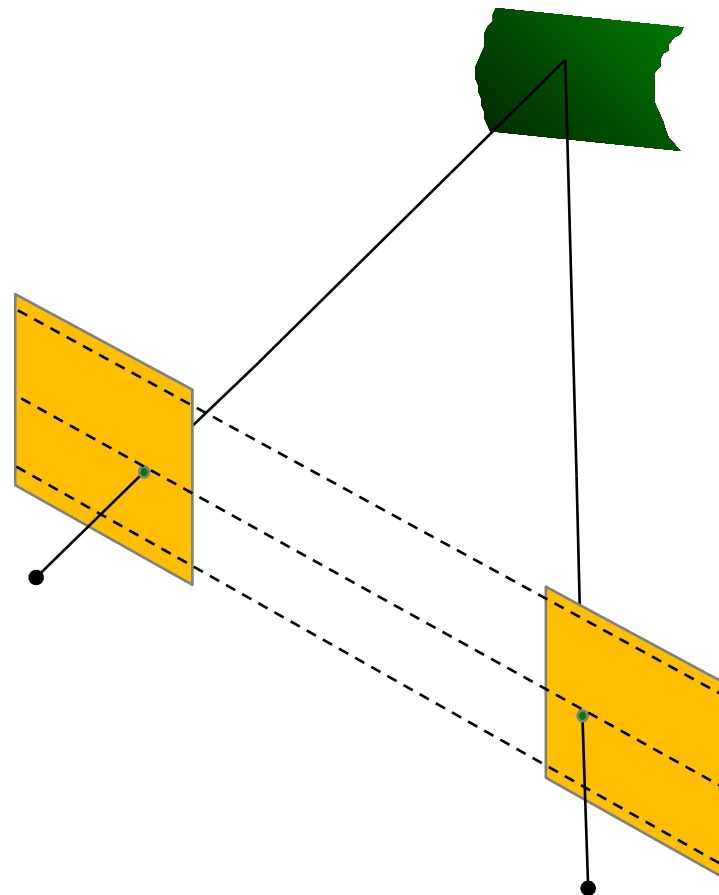
Stereo Image Rectification

- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R
- 3) Rotate both images by R_{rect}
- 4) Scale images



Stereo Image Rectification

- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R
- 3) Rotate both images by R_{rect}
- 4) Scale images



Actually not so simple...

- 1) Estimate E using the 8 point algorithm
- 2) Estimate the epipole e (solve $Ee=0$)
- 3) Build R_{rect} from e
- 4) Decompose E into R and T
- 5) Set $R1=R_{\text{rect}}$ and $R2 = RR_{\text{rect}}$
- 6) Rotate each left camera point $x' \sim Hx$ where $H = KR1$
- 7) Alter the focal length (inside K) to keep as much points within the original image size
- 8) Repeat 6 and 7 for right camera points using $R2$



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Epipolar Geometry

Question Time!





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Epipolar Geometry



GEOMETRIA EPIPOLARE

- Why stereo?
- Epipolar constraints
- Essential and Fundamental matrices
- Stereo Images rectification

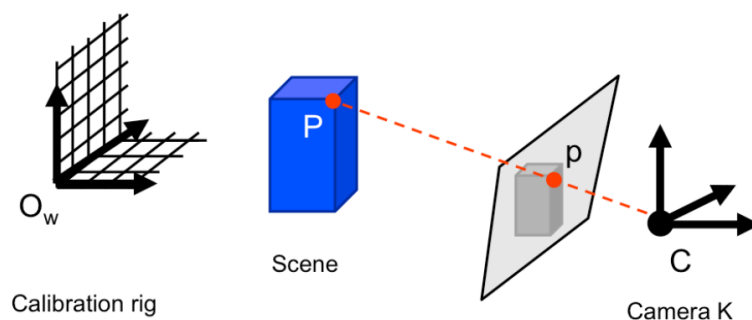
Prima di tutto capiamo perché lo stereo (e questa è facile), formalizzeremo la geometria che ci sta dietro e capiremo cosa sono e come si ricavano due matrici che legano due immagini stereo. Infine parleremo di rettificazione



- [FP] D. A. Forsyth and J. Ponce. Computer Vision: A Modern Approach (2nd Edition). Prentice Hall, 2011.
- [HZ] R. Hartley and A. Zisserman. Multiple View Geometry in Computer Vision. Cambridge University Press, 2003.
- **CS231A · Computer Vision: from 3D reconstruction to recognition, Prof. Silvio Savarese – Stanford University**

Credits for many images/equations: Silvio Savarese

Come sempre un po' di riferimenti dei corsi da cui viene parte del materiale



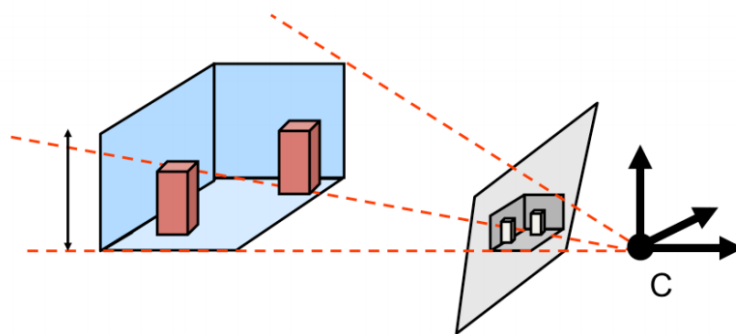
Calibration rig + rig position



We can compute K and RT

In the previous lectures, we have seen how to compute the intrinsic and extrinsic parameters of the camera using 1 or more view(s) – this is the camera calibration procedure.

We have also derived equations for estimating the intrinsics of the camera (i.e., the matrix K) from just one image (geometry of the vanishing points and lines) as well as for estimating some of the properties of the 3D world such as surface primitives, ground planes, etc..., given that some prior knowledge about the world is available (for instance, a building façade is orthogonal to the ground plane).



Vanishing lines and points + orthogonal planes



We can compute K

In the previous lectures, we have seen how to compute the intrinsic and extrinsic parameters of the camera using 1 or more view(s) – this is the camera calibration procedure.

We have also derived equations for estimating the intrinsics of the camera (i.e., the matrix K) from just one image (geometry of the vanishing points and lines) as well as for estimating some of the properties of the 3D world such as surface primitives, ground planes, etc..., given that some prior knowledge about the world is available (for instance, a building façade is orthogonal to the ground plane).



Why it is **soooo** difficult?



Line of Sight: multiple potential correspondences for p in the 3D world..

However, it is in general not possible to recover the structure of the 3D world (e.g., the blue box; the point P) from just one image. This is due to the intrinsic ambiguity of the 3D to 2D mapping. **Any point along the ray from the camera center C to a point p in the image (line of sight) is a possible candidate for the actual 3D point P .**



Questa slide l'ha inserita il precedente docente. Mi piace molto e l'ho tenuta. Viene dal film *Notorious* che è del '46. Hitchcock amava mettere in primo piano elementi fondamentali alla trama (nella tazzina c'è del veleno). Ma aveva un piccolo problemillo, le telecamere all'epoca (in realtà è vero parzialmente anche adesso) avevano sicuramente dei problemi a mettere a fuoco elementi così vicini.

Soluzione? Quella tazzina non è affatto vicina. E come è possibile, si capisce bene che lo è... E invece no, ci facciamo ingannare dalla conoscenza a priori che abbiamo della tazzina o meglio delle sue dimensioni. Quella tazzina è lontana e pure enorme.

Ma da una sola immagine o meglio da visione mono, non è possibile capirlo.

Without stereo...

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Qui l'effetto è ancora più evidente e anzi viene sfruttato per la comicità del film stesso (Top Secret!, 1984)

Without stereo...

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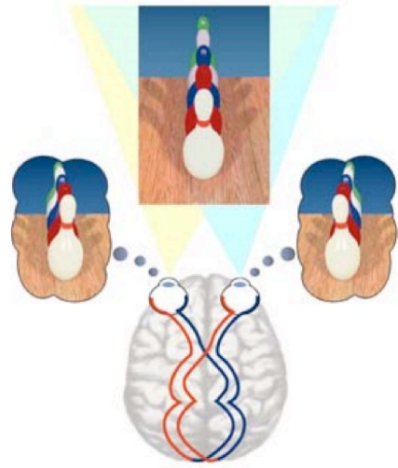


La Monarca ha circa 120.000 occhi

Le capesante (rock scallop) possono averne un centinaio

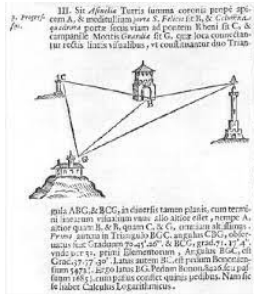
Il lucertolo ha il cosiddetto occhio parietale

L'unico bestio monoculare è il Copepoda, crostacei da 1-2 mm...



So, what if there are two cameras observing the object? As we will shortly see, it turns out that it is quite helpful! Humans and most other life forms have evolved a pair of eyes so that the world around them can be perceived in 3D. Imagine how it would have been if we were only capable of perceiving 2D images!

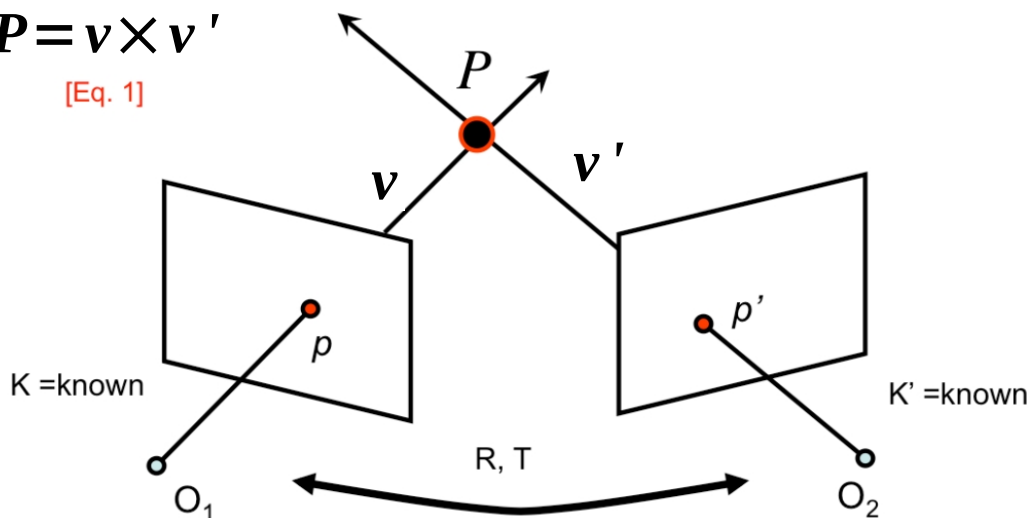
Triangulation



Triangolare significa suddividere in triangoli, procedimento simile a quello del P3P insomma

$$P = v \times v'$$

[Eq. 1]



Supponiamo di avere due telecamere che inquadrano la scena. Anche se fossero dello stesso tipo e modello ci sta che abbiano intrinseci lievemente differenti $\rightarrow$ K e K'

Consideriamo un punto del mondo P osservato dalle due telecamere. Le sue proiezioni nei due piani immagine saranno p e p' . MA non sappiamo dove si trovi (o meglio non ancora)

Le due telecamere vedono P tramite due linee di vista v e v' . Come visto in lezioni precedenti, P lo posso stimare come prodotto vettoriale di v e v' . Dato che v e v' li posso calcolare grazie a K e K' è ragionevole capire come grazie allo stereo io possa descrivere il mondo 3D ovvero stimare nuovamente il 3D.

In realtà c'è qualche passaggio che ci frega:

1. p e p' in generale non li conosco
2. vero che $P = v \times v'$ ma v e v' li devo esprimere nello stesso sistema di riferimento

Insomma c'è qualcosina su cui lavorare...

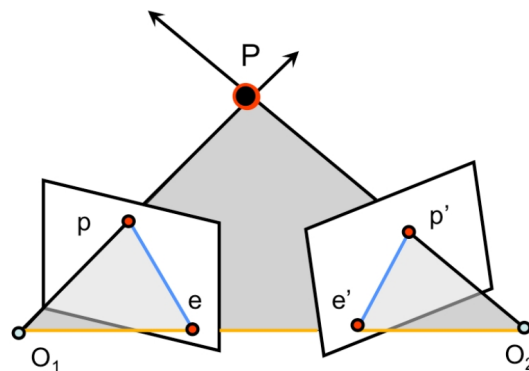


- **Correspondances:** given a p in one image find p' in the other image
- **Camera geometry:** given a set of correspondences find camera parameters
- **Scene Geometry:** given a set of correspondences and parameters of both cameras reconstruct the 3D scene

Quindi i problemi aperti sono:

1. dato un p in una immagine trovare il relativo p' nell'altra
2. come posso ricavare i parametri complessivi che legano le due telecamere stereo?
3. dati i parametri delle telecamere e diverse corrispondenze p/p' calcolarmi il 3D

L'(1) lo vediamo nel prossimo pacco di slide, il (2) ci sono slide apposite, diciamo che per ora focalizziamoci solo sul (3)



- O₁-O₂-P: epipolar plane
- O₁-O₂: baseline
- pe & p'e': epipolar lines (all epipolar lines meet in e or e')
- e & e': epipoles
 - Intersection of baseline with image planes
 - Projection of O₁&O₂

Un po' di definizioni

La linea che connette i due pin hole ovvero i due centri ottici è la linea epipolare

Chiamiamo il piano determinato dai centri ottici e da P come piano epipolare.

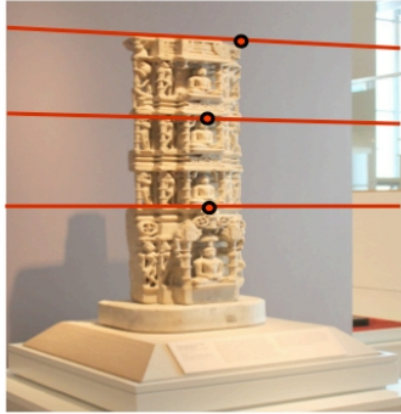
Questo piano interseca i piani immagine in due rette che sono le rette epipolari

e ed e' sono i punti di intersezione tra la baseline e i piani immagine

Al variare di P cambieranno:

- il piano epipolare
- le proiezioni p e p'
- le rette epipolari

NON cambieranno e ed e'. Quindi tutte le rette epipolari di ciascun piano immagine si incroceranno (al variare di P) comunque in e o e'



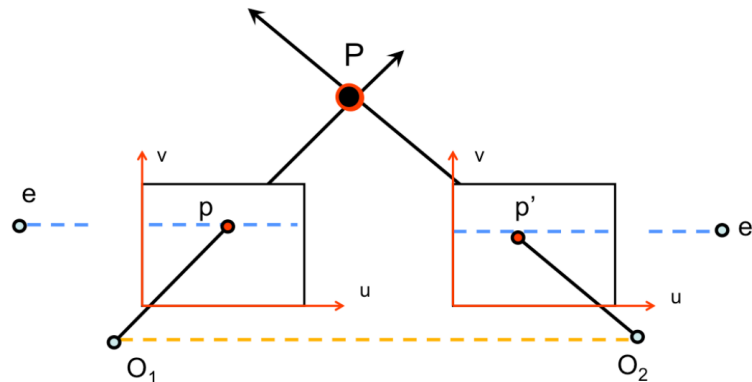
I pallini rappresentano le proiezioni di punti in immagini stereo e le associate linee epipolari (e ed e' sarebbero un po' troppo fuori per disegnarli).
Si può comunque notare come non siano parallele...



Caso in cui il non parallelismo è ancor più accentuato

Esistono condizioni in cui le rette epipolari sono parallele? Teoricamente sì

Special case: parallel image planes

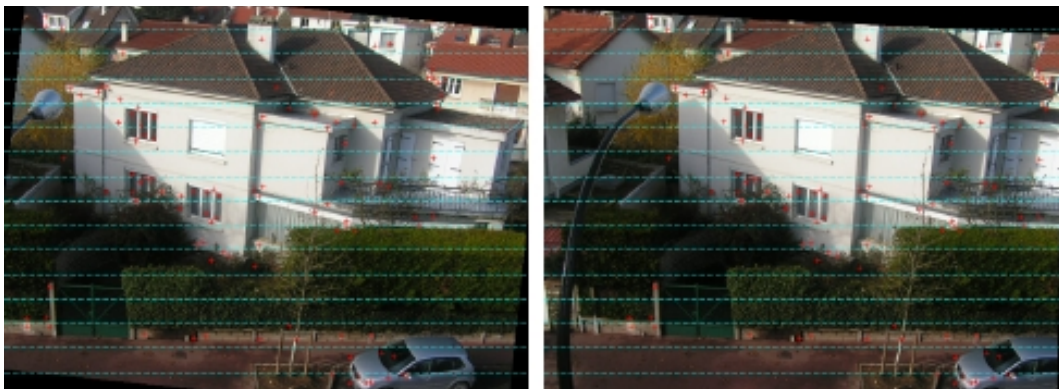


- Epipoles are at infinity
- Also intersection between baseline and image plane are at infinity
- Epipolar lines are parallel to u axis

Nel caso speciale in cui i piani immagini siano paralleli tra di loro effettivamente gli epipoli e ed e' vanno all'infinito e le linee epipolari sono parallele tra di loro ma anche (in questo caso specifico) a O_1O_2 .

Ad essere pignoli qui è disegnata la situazione in cui i piani immagine sono anche coplanari e orientati di modo che l'asse delle u sia parallelo alla baseline.

È una situazione ideale e vedremo che ambisco ad ottenerla



- Example of two images whose planes are parallel to each other
- Epipolar lines are parallel to each other

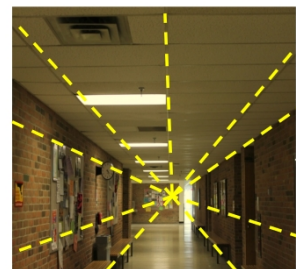
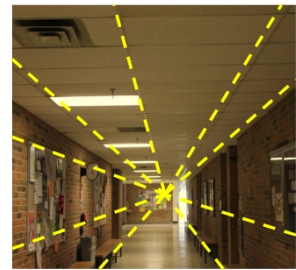
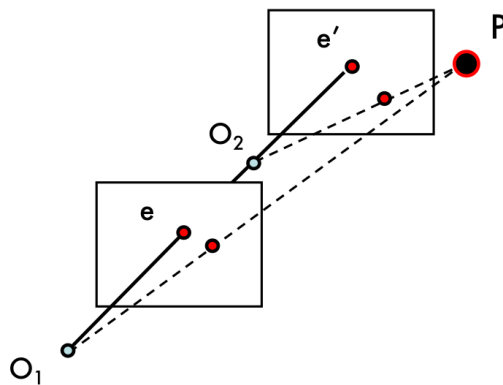
Ecco un esempio della situazione precedentemente descritta.

Attenzione, difficilmente riesco ad ottenere una situazione in cui i piani immagine siano realmente paralleli e coplanari. Posso essere preciso quanto voglio a fissare e orientare una coppia di telecamere, ma ben difficilmente otterrò tale situazione. Tenendo anche conto che la telecamera stessa sarà giocoforza “imprecisa” già a livello costruttivo.

Queste due immagini sono state ottenute tramite una procedura di **rettificazione**. In pratica proietto i due piani immagine su un piano comune e oriento gli assi di modo che siano paralleli.

Qualcuno ha parlato di omografia?

Special case: forward translation

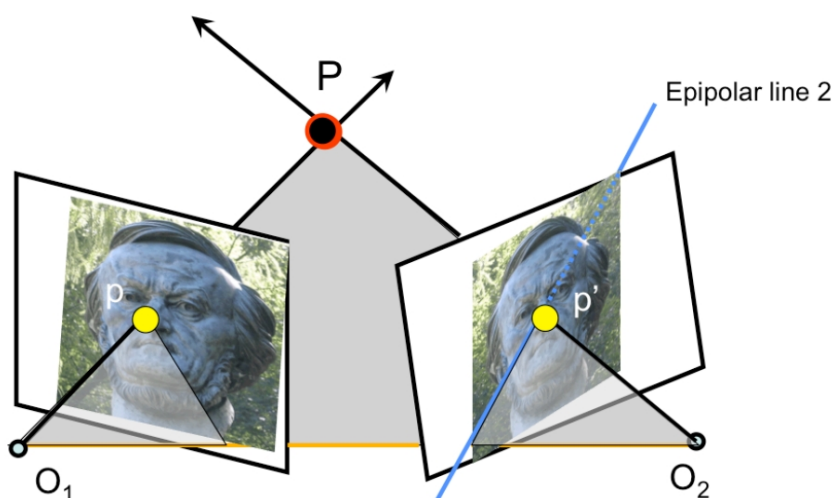


Giusato una introduzione al motion stereo...

Non ho stereo solo acquisendo allo stesso istante da due telecamere differenti ma anche acquisendo in momenti differenti dalla stessa telecamera.

Ad esempio frame successivi di una telecamera che viene spostata.

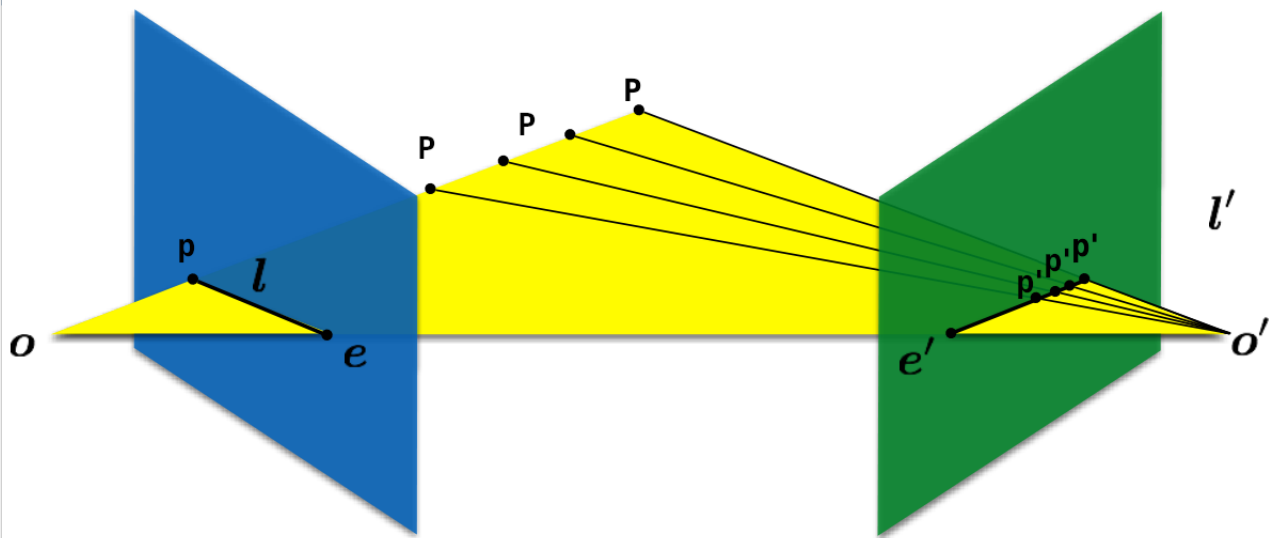
Se il movimento è traslatorio senza rotazioni, gli e saranno nello stesso punto in entrambe le immagini. E si chiama FOE ovvero Focus of Expansion (d'altro canto tutto si muove lungo le linee epipolari che partono da lì... guardate ad esempio i poster)



Perché le linee epipolari sono importanti?

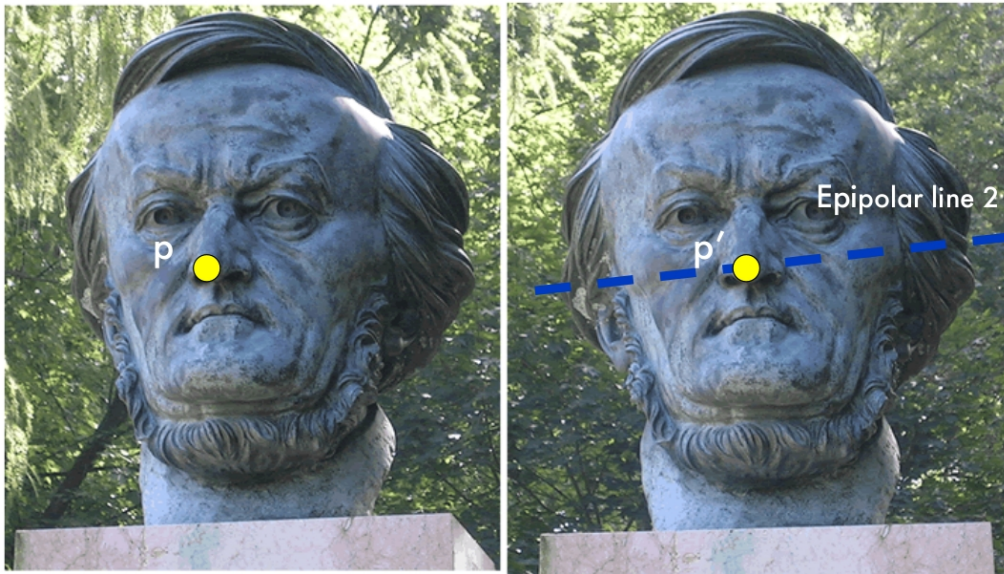
Perché dato un p è dove devo cercare il p' (visto che io ho le immagini ma non so dove effettivamente sia P). Quindi per i punti dell'immagine sinistra, io posso cercare la relativa corrispondenza nell'immagine destra ma SENZA cercare ovunque, ma solo sulle linee epipolari.

Si noti come il piano epipolare e quindi le linee epipolari io le posso ricavare anche senza conoscere P e p' , mi bastano p , O_1 e O_2

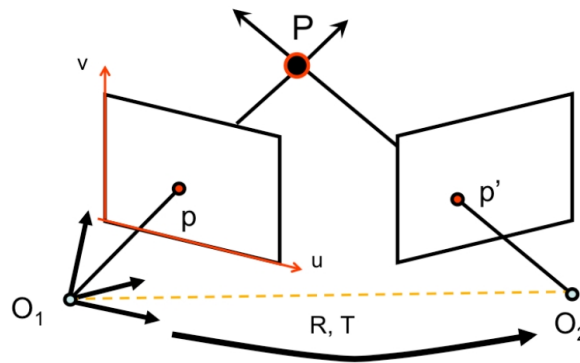


Credits: Ioannis Gkioulekas

Dato p e senza sapere dove sia P comunque riesco a vincolare la posizione di p' che sarà sulla relativa linea epipolare
E questo è un buon esempio.



Dato p e senza sapere dove sia P comunque riesco a vincolare la posizione di p' che sarà sulla relativa linea epipolare



$$M = K \begin{bmatrix} I & 0 \end{bmatrix}$$

$$M P = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = p \quad [\text{Eq. 3}]$$

$$M' = K' \begin{bmatrix} R^T & -R^T T \end{bmatrix}$$

$$M' P = \begin{bmatrix} u' \\ v' \\ 1 \end{bmatrix} = p' \quad [\text{Eq. 4}]$$

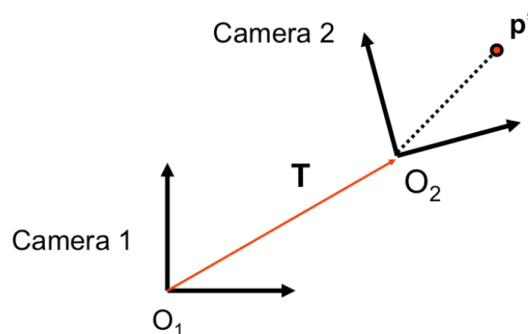
In questo caso pongo il sistema di riferimento sulla telecamera SX. M' lo derivo come K' per rototraslazione come di solito. Solo che qui la rototraslazione è verso l'altra telecamera.

In questo caso la matrice di trasformazione M per la telecamera destra è quella usuale. La matrice M' è quella che mi genera i p' ma con coordinate rispetto a O1. Si può dimostrare che è quella indicata.

Basta invertire la matrice 4x4

$$\begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^T & -R^T T \\ 0 & 1 \end{bmatrix}$$

Di fatto è una omografia... e ricordiamoci che per le rotazioni vale che $R^T = R^{-1}$



$$p'_{O_1} = R p'_{O_2} + T$$

$$p'_{O_2} = R^T (p'_{O_1} - T)$$

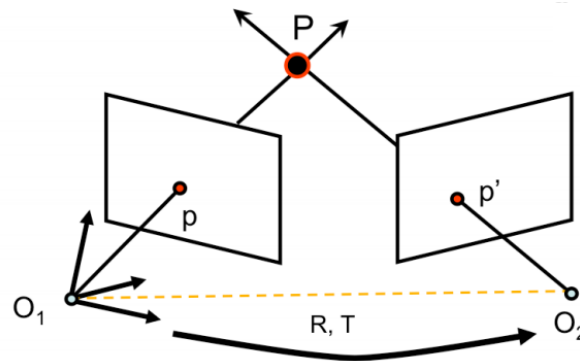
- T is O_2 in the O_1 reference system
 - Actual translation
- R is the rotation matrix such as **free vector** p' for O_2 is $R p'$ for O_1
 - Rotation limited to epipolar plane

Va comunque detto che non è esattamente una rototraslazione completa. Rammento che le matrici di rotazione sono ortogonali e che quindi $R^{-1} = R^T$. Assume T is the coordinate vector of the translation $O_1 O_2$ separating the two coordinate systems. T gives the translation between the two cameras and R gives the relative rotation. Using T , we can determine the coordinates of the 2nd camera center O_2 . Since the 1st camera center is assumed to be at the origin (of the world reference system), then the center (O_2) of the 2nd camera will be at T . R is the rotation matrix such that a free vector p' in the camera 2 is equal to $R p'$ in camera 1.

- Is a camera where $K = I$
- Not a real camera but a simplified model that can ease to understand epipolar relations
- Actually used when we know the K and K' of real cameras
 - i.e. the cameras are already calibrated

$$K_{canonical} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad p = Kp_c$$

Quindi $f \cdot \text{fattore} = 1$, no skew e pixel quadrati. Chiaro che non esiste. Ma se già conosciamo i due K ci aiuta. Quindi inizialmente facciamo questa assunzione per poi generalizzare dopo.

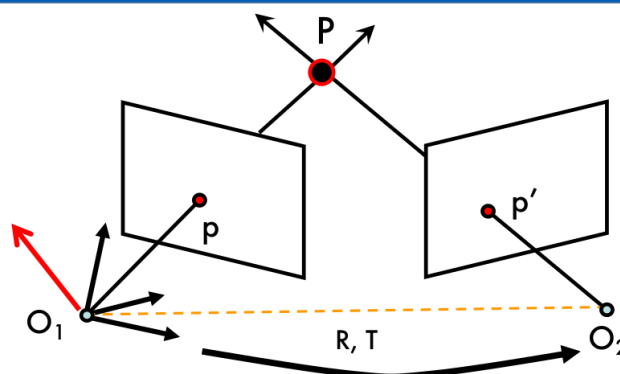


$$K_{canonical} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = K \begin{bmatrix} I & 0 \end{bmatrix} \quad \boxed{K = K' \text{ are known (canonical cameras)}} \quad M' = K' \begin{bmatrix} R^T & -R^T T \end{bmatrix}$$

$\downarrow$ $M = \begin{bmatrix} I & 0 \end{bmatrix}$ [Eq. 5] $\downarrow$ $M' = \begin{bmatrix} R^T & -R^T T \end{bmatrix}$ [Eq. 6]

Se la telecamera è quella canonica nel calcolo di M e M' spariscono i K di fatto



p' in first camera reference system is $= R p' + T$
 $T \times ((R p') + T) = T \times (R p')$ is perpendicular to epipolar plane

$$\rightarrow p^T \cdot [T \times (R p')] = 0 \quad \text{[Eq. 7]}$$

Abbiamo detto che p' lo vedo come $Rp' + T$ nel sistema di riferimento della telecamera sx.

Se moltiplico vettorialmente per T e distribuisco la moltiplicazione (sì, anche il prodotto vettoriale è distribubile) $T \times T$ si annulla e rimane quindi solo $T \times R p'$

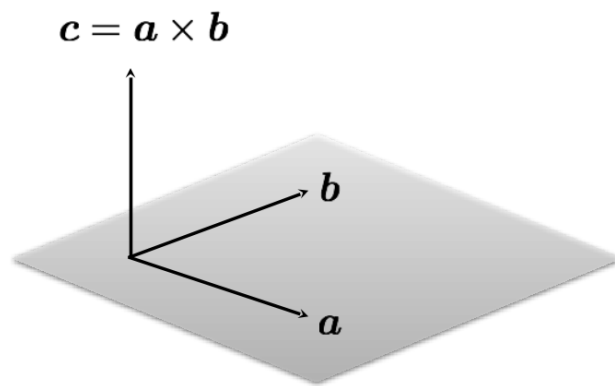
Sarà un vettore ortogonale al piano epipolare visto che Rp' e T giacciono entrambi su quel piano (T è il vettore che porta da $O1$ a $O2$ e Rp' il vettore che porta da $O1$ a p' e tutti questi punti giacciono sul piano epipolare).

Ma quindi il prodotto scalare di p (che anche lui giace sul piano epipolare) e quello è sicuramente 0 (sono vettori ortogonali tra di loro)!

Nota: quel punto della moltiplicazione scalare ce lo tiriamo dietro. Perché lo evidenziamo così mentre per altre moltiplicazioni del tutto equivalenti no? Ma per stressare il fatto che ho la moltiplicazione tra un punto e una linea

Cross product recall

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$$\mathbf{c} \cdot \mathbf{a} = 0$$

$$\mathbf{c} \cdot \mathbf{b} = 0$$

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}$$

cross product of two vectors
in the same direction is zero

$$\mathbf{a} \times \mathbf{a} = 0$$

remember this!!!

Credits: Ioannis Gkioulekas

Un piccolo recap di moltiplicazioni di vettori, sia scalari che vettoriali, a scopo di riassunto e per introdurre quanto segue

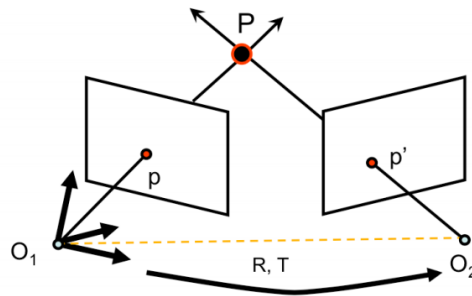


$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & -a_x & 0 \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = [\mathbf{a}_x] \mathbf{b}$$

$$\mathbf{a} \times \mathbf{b} = [\mathbf{a}_x] \mathbf{b}$$

$$\mathbf{T} \times (\mathbf{R} \mathbf{p}') = [\mathbf{T}_x] (\mathbf{R} \mathbf{p}') = ([\mathbf{T}_x] \mathbf{R}) \mathbf{p}'$$

Esiste il modo di tradurre un prodotto vettoriale in prodotto matriciale? Sì se il primo moltiplicando lo esprimiamo come quella matrice. Matrice che nel seguito indicheremo con la notazione $[\text{<nome vettore>}_x]$

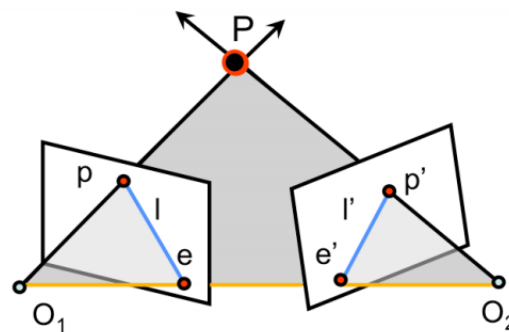


$$p^T \cdot [T \times (Rp')] = 0$$

$$p^T \cdot \underbrace{\left([T_x] R \right)}_E p' = 0$$

- E is the **Essential Matrix**

Quindi posso riscrivere la relazione che lega p e p' eliminando il prodotto vettoriale. Anzi, raccogliendo meglio le moltiplicazioni di fatto ottengo una matrice che mi lega p e p', la cosiddetta matrice essenziale



$$p^T \cdot E p' = 0$$

- $l = E p' \rightarrow$ epipolar line for p
- $l' = E^T p \rightarrow$ epipolar line for p'
- $E e' = 0 \quad E^T e = 0 \rightarrow$ namely epipoles are solution for homogeneous equation
- E is a 3×3 matrix with 5 DOF
- E is singular and its rank is 2

NON È UNA OMOGRAFIA CHE MAPPA PUNTI A PUNTI SU UN ALTRO PIANO, MA MAPPA UN PUNTO A UNA LINEA

Per $l = E p'$: ricordiamo che se un punto giace su una linea vale che il prodotto scalare tra p^T e linea vale 0, da $p^T E p' = 0$ se ne deduce che $E p'$ lo posso vedere come una linea. Ritornando a come è stata ricavata se ne deduce il perché sia esattamente la linea epipolare.

$E e' = 0$: tutte le linee epipolari passano per gli epipoli. Quindi per qualunque p con $E^T p$ otteniamo una linea che passa per e' ovvero $e'^T E^T p = 0$, ma se deve valere per qualunque p va da se che $e'^T E^T$ deve essere lui 0

Perché 5 gradi di libertà e non 6 (3 rotazioni + 3 traslazioni)? In realtà ricordiamoci che non è una rototraslazione completa. Il determinante=0 a 0 mi riduce di un grado. Rango $2 < 3$ implica che il determinante sia 0



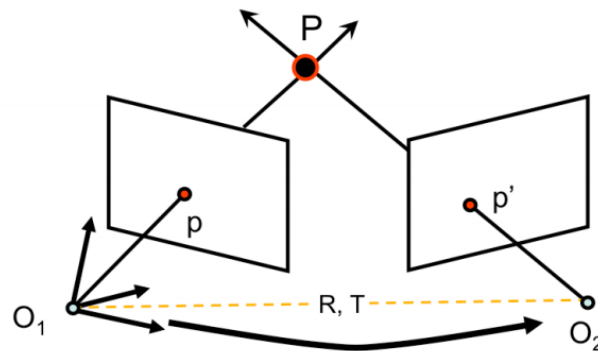
$$\mathbf{p}^T \mathbf{E} \mathbf{p}' = 0 \quad \text{Longuet-Higgins Equation}$$

- Usually we do not use ideal cameras
- Is then E so useful?
- Yes if we think about normalized camera coordinates and not image coordinates
 - For image coordinates we need to know K

$$\mathbf{p} = \mathbf{K} \mathbf{p}_c \quad \mathbf{p}_c = \mathbf{K}^{-1} \mathbf{p}$$

Se trascuriamo skew e scala nella mia matrice rimane solo f se divido per f di fatto normalizzo.

Epipolar constraint (non ideal camera)



$$M = K \begin{bmatrix} I & 0 \end{bmatrix}$$

$$p_c = K^{-1} p \quad \text{[Eq. 11]}$$

$$M' = K' \begin{bmatrix} R^T & -R^T T \end{bmatrix}$$

$$p'_c = K'^{-1} p' \quad \text{[Eq. 12]}$$

Riesco quindi a convertire!

p_c e p'_c sono quindi le proiezioni con camera canonica, p e p' quelle di una camera non canonica.

MA perché converto da p a p_c ? Beh p lo conosco e per applicare la matrice essenziale mi serve p_c

Now, let's derive a similar relationship between p and p' when the K s are no longer canonical. So suppose that the cameras are not canonical and let K and K' be the (unknown) camera matrices for camera 1 and 2, respectively. In this case, we can modify the previous derivation to arrive to a slightly different constraint which we define in the next slide. Let p_c and p'_c be the projections of P to the corresponding camera images that we derived by assuming that the camera matrices are canonical. Let p and p' be the projections of P to the corresponding camera images by assuming that the cameras are not canonical with K and K' as defined above. In this case Eq. 11 and 12 are satisfied.

$$p_c = K^{-1} p$$

[Eq. 11]

$$p'_c = K'^{-1} p'$$

[Eq. 12]

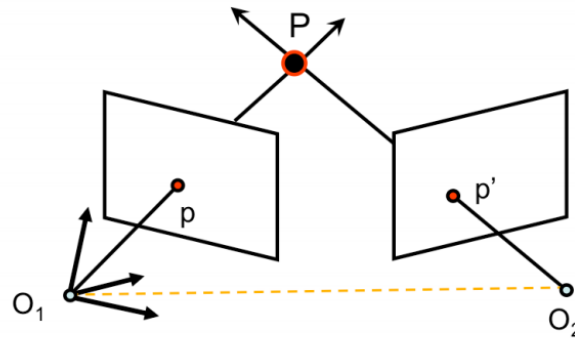
[Eq.9]

$$p_c^T \cdot [T_{\times}] \cdot R p'_c = 0 \rightarrow (K^{-1} p)^T \cdot [T_{\times}] \cdot R K'^{-1} p' = 0$$

$$p^T [K^{-T} \cdot [T_{\times}] \cdot R K'^{-1}] p' = 0 \rightarrow p^T F p' = 0 \quad \text{[Eq. 13]}$$

p_c e p'_c sono quindi le proiezioni con camera canonica, P e p' quelle di una camera non canonica.

Using Eq. 11 and 12, we replace the expressions of p_c and p'_c in Eq.9 and, by follow the above derivation, we obtain a new epipolar constraint, as given in equation 13.



[Eq. 13]

$$p^T F p' = 0$$

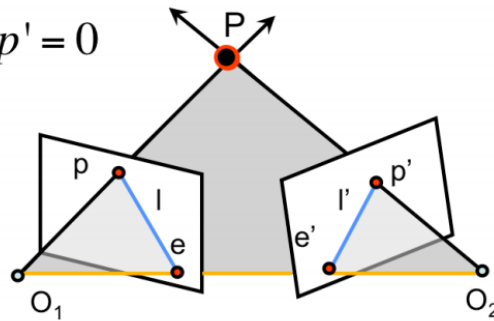
$$F = K^{-T} \cdot [T_x] \cdot R K'^{-1}$$

[Eq. 14]

- $F \rightarrow$ Fundamental Matrix (Faugeras and Luong 1992)

Using Eq. 11 and 12, we replace the expressions of p_c and p'_c in Eq.9 and, by following the above derivation, we obtain a new epipolar constraint, as given in equation 13.

$$p^T \cdot F \cdot p' = 0$$



- $l = Fp' \rightarrow$ epipolar line for p
- $l' = F^T p \rightarrow$ epipolar line for p'
- $Fe' = 0 \quad F^T e = 0 \rightarrow$ namely epipoles are solution for homogeneous equation
- F is a 3×3 matrix with 7 DOF
- F is singular and its rank is 2

Perché 7 e non 9? Le coordinate omogenee fanno sì che la scala mi arrivi nel risultato e il $\det(F) = 0$ mi riduce di 1 altro vincolo.

The fundamental matrix is also useful to compute the epipolar lines associated with p and p' , when the camera matrices K , K' , and the transformation R and T are unknown. These relationships are equivalent to the ones we introduced for the essential matrix. The key difference is that F has 7 degrees of freedom (instead of 5)

Sui 7, ci sono 9 elementi ma la scala non importa e $\det(F) = 0$ ce ne rimuove un altro.



$$p_c^T E p'_c = 0 \quad p^T F p' = 0$$

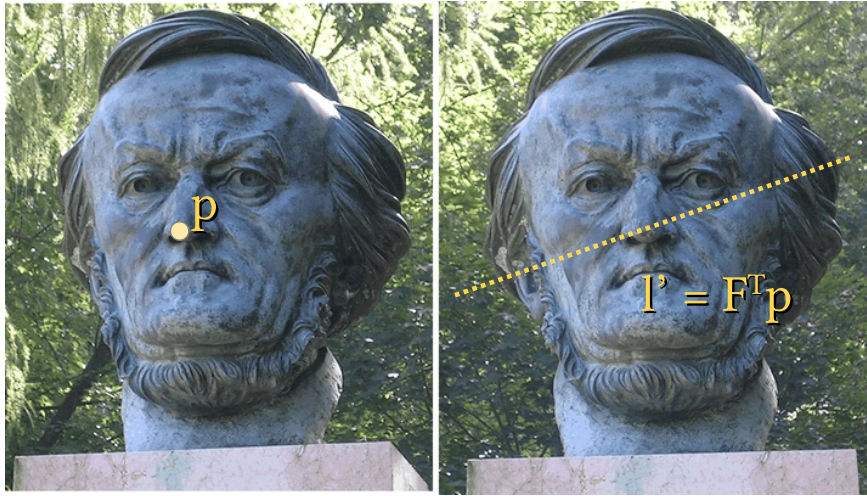
$$p_c = K^{-1} p \quad p'_c = K'^{-1} p'$$

$$p^T K^{-T} E K^{-1} p' = 0$$

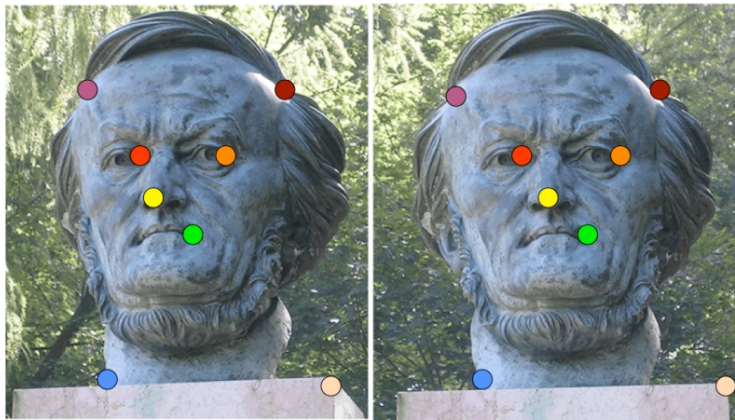
$$K^{-T} E K^{-1} = F$$

In maniera analoga si può ricavare $E = K'^T F K$ ma questo è meno utile

Why we want F (or E)?



Noi vogliamo F in quanto nelle immagini reali dato un p sappiamo dove cercarlo nell'altra immagine ovvero sulla linea epipolare. Ed è questa quella che ci interessa.



- Can we estimate F without knowing nothing about K , R , T ?
- Yes $\rightarrow$ 8-point algorithm

Se conosco le K con lo stesso metodo ricavo la E che mi basta visto



- Why those matrices are important?
- They give us a relation between two stereo pairs even without knowing anything about their calibration
 - The 8 points approach can be used to estimate them
- In the next lesson we are going to see how this, limited, knowledge anyway allows us to reconstruct a 3D scene

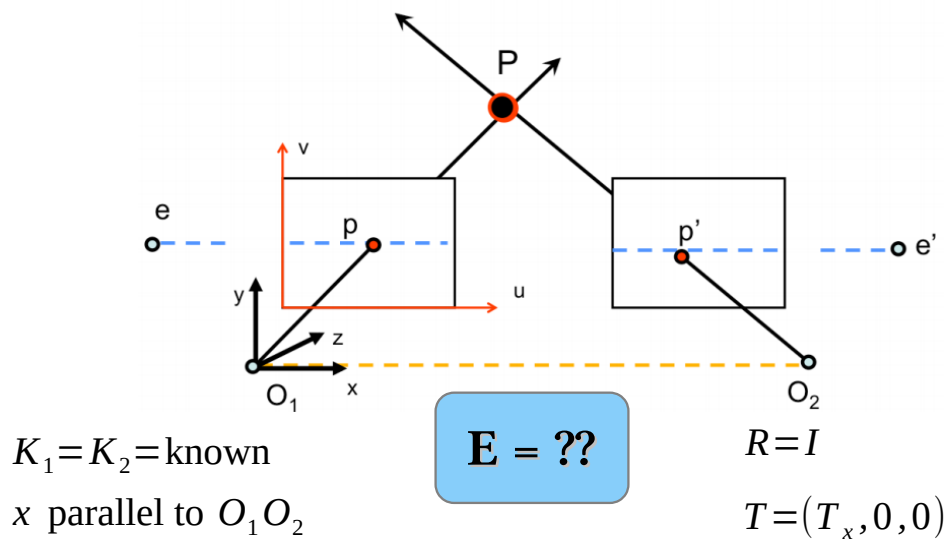


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Special Case Parallel Image Planes

SINGLE VIEW RECONSTRUCTION

Special case: parallel image planes



In realtà basta siano paralleli non necessariamente sullo stesso piano.

Let us try to compute the essential matrix E in the case of parallel image planes. We assume that the two cameras have the same K , and that K is known. Since the image planes are parallel, there is no relative rotation between the two cameras. So, $R = I$. There is only a translation T , along the x -axis. So, $T = (T, 0, 0)$.

With this information, we set out to find E ...

$$E = [T_x] \cdot R$$

$$R = I \quad T = [T_x \quad 0 \quad 0] = [T \quad 0 \quad 0]$$

$$E = \begin{bmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & T_x & 0 \end{bmatrix} \cdot I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -T \\ 0 & T & 0 \end{bmatrix}$$

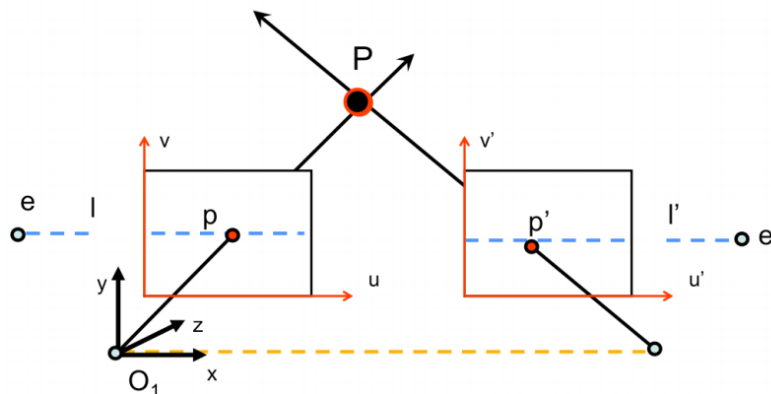
In pratica solo T_x è diverso da zero e lo chiamiamo per semplicità T

In realtà basta siano paralleli non necessariamente sullo stesso piano.

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With this information, we set out to find E ...

Special case: parallel image planes



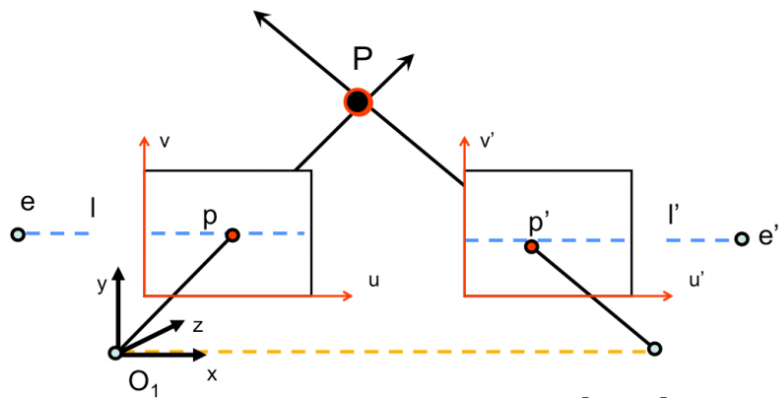
- Epipolar lines are horizontal

$$l = E p' = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -T \\ 0 & T & 0 \end{bmatrix} \begin{bmatrix} u' \\ v' \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -T \\ T v' \end{bmatrix} \text{ horizontal!}$$

Orizzontale in quanto non dipende dalla u

Once E is known, we can find the directions of the epipolar lines associated with points in the image planes. Let us compute the direction of the epipolar line l associated with point p' . The direction of the epipolar line $l = E p'$. We can see that the direction of l is horizontal, as we would expect to see in this case. We can perform $E p$ to compute l' , which can also be verified to be horizontal.

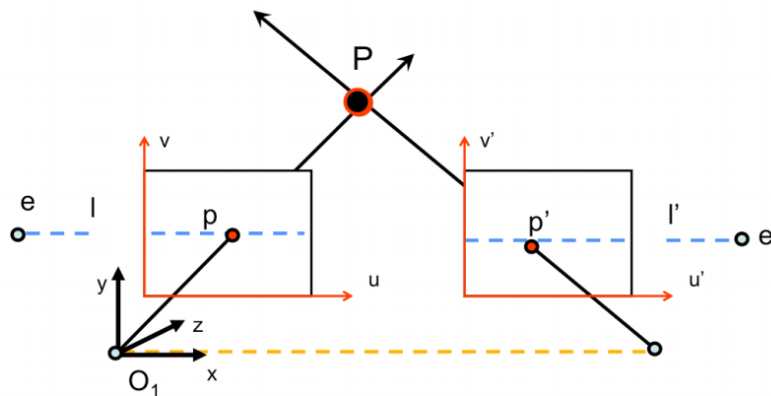
Special case: parallel image planes



- How p & p' are related?

$$p^T \cdot E \cdot p' = 0$$

Special case: parallel image planes

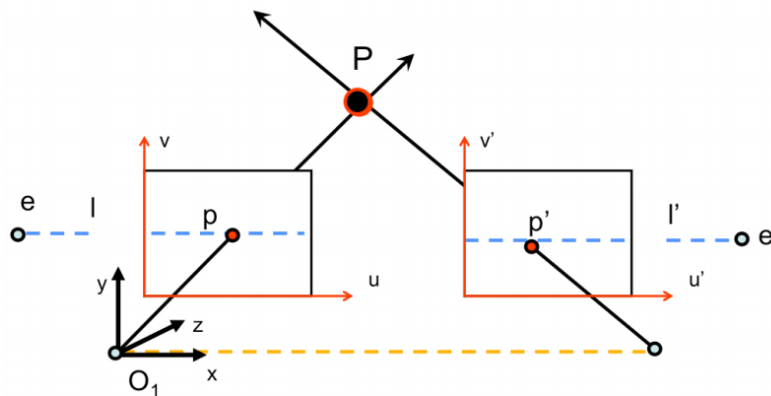


- How p & p' are related?

$$(u \ v \ 1) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -T \\ 0 & T & 0 \end{bmatrix} \begin{pmatrix} u' \\ v' \\ 1 \end{pmatrix} = 0 \Rightarrow (u \ v \ 1) \begin{pmatrix} 0 \\ -T \\ Tv' \end{pmatrix} = 0 \Rightarrow Tv = Tv' \Rightarrow v = v'$$

We can use the Epipolar constraint to relate the y coordinates of p and p' . As the slide shows, $v = v'$ which implies that p and p' share the same v coordinate as expected.

Special case: parallel image planes



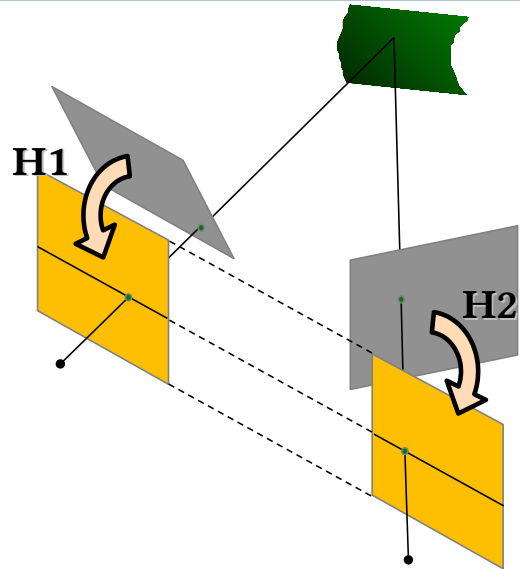
- $v=v'$ means the same line!
- The search for p' is simpler...

True for canonical cameras!

Also when $K=K'$!

We can use the Epipolar constraint to relate the y coordinates of p and p' . As the slide shows, $v = v'$ which implies that p and p' share the same v coordinate as expected.

- Parallel image planes are better to handle for stereovision
- Simply accurately align cameras
 - Good idea, but it does not work!
 - Precision, noise, thermal effects...
- Reproject camera planes onto a common plane parallel to the line between camera centers



Rectification is the process of making two given images parallel. It is useful because there is a straightforward relationship between corresponding points (they share the same v coordinate) when two images are rectified



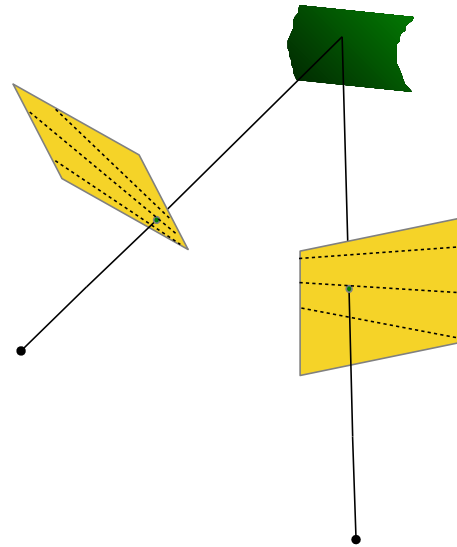
- Assuming $K=K'$ as known
 - 1) Compute $E \rightarrow$ we get R
 - 2) Rotate right image by R
 - 3) Rotate both images to move e and e' to ∞
 - We need to compute a R_{rect}
 - 4) Adjuste the scale

Questi sono i passi principali

Gli epipoli sono intersezione linee epipolari e linea O_1O_2



1) Compute $E \rightarrow$ we get R

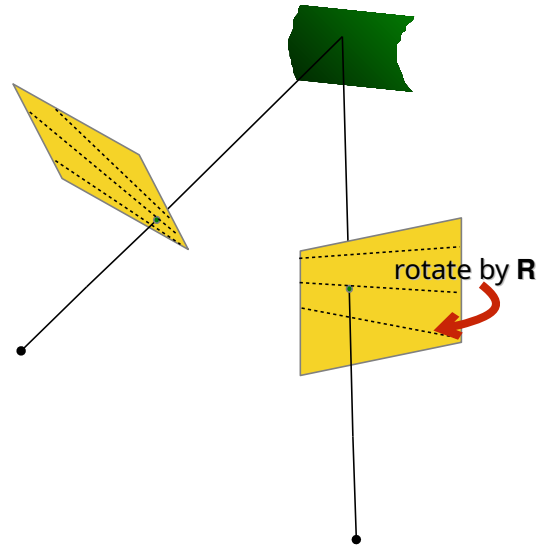


Credits: Ioannis Gkioulekas

Con il metodo degli 8 punti ricavo la matrice E da cui ricavo la rotazione tra i due piani immagine R . Con R ruoto la seconda immagine di modo da portarla sullo stesso piano della prima



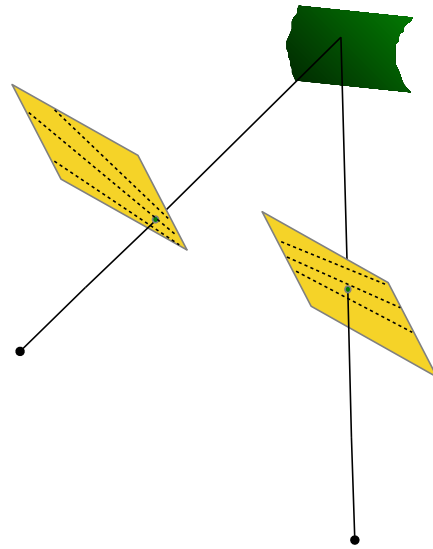
- 1) Compute $E \rightarrow$ we get R
- 2) Rotate left image by R



Credits: Ioannis Gkioulekas



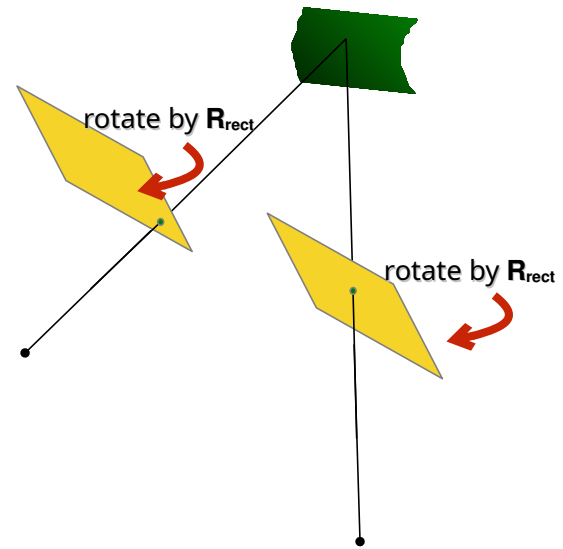
- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R



Credits: Ioannis Gkioulekas



- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R
- 3) Rotate both images by R_{rect}

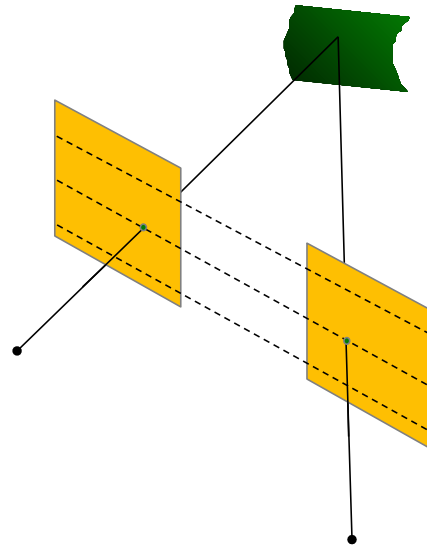


Credits: Ioannis Gkioulekas

Ora sono sullo stesso piano ma gli assi non sono ancora paralleli. Calcolo un R_{rect} che mi manda gli epipoli all'infinito e lo uso



- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R
- 3) Rotate both images by R_{rect}

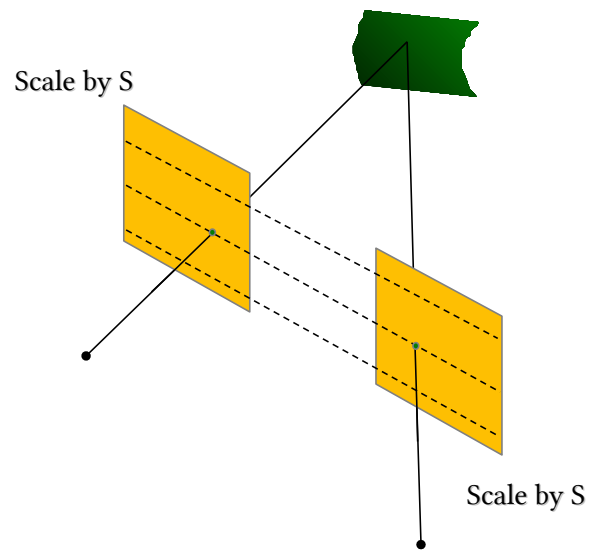


Credits: Ioannis Gkioulekas

Qui sono rappresentate come rettangolari ma di fatto non lo sono visto che una l'ho proiettata su altro piano. Oltretutto potrei perderne dei pezzi. Meglio scalarle (che poi in pratica significa proiettarle su un piano più vicino/lontano)



- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R
- 3) Rotate both images by R_{rect}
- 4) Scale images

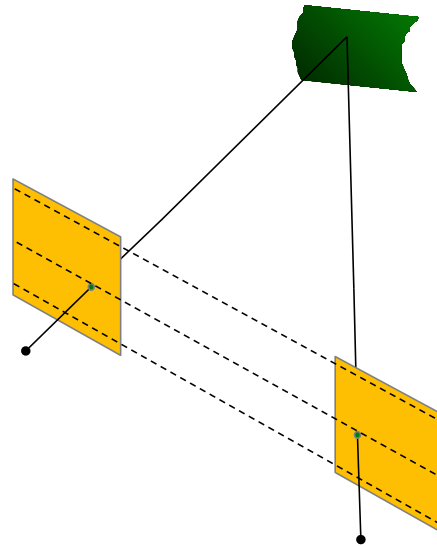


Credits: Ioannis Gkioulekas

Ottimizzo la parte di immagine contenuta effettivamente nell'immagine finale (che avrà delle dimensioni)



- 1) Compute $E \rightarrow$ we get R
- 2) Rotate right image by R
- 3) Rotate both images by R_{rect}
- 4) Scale images



Credits: Ioannis Gkioulekas

Gli epipoli sono intersezione linee epipolari e linea O_1O_2

Actually not so simple...

- 1) Estimate E using the 8 point algorithm
- 2) Estimate the epipole e (solve $Ee=0$)
- 3) Build R_{rect} from e
- 4) Decompose E into R and T
- 5) Set $R_1=R_{\text{rect}}$ and $R_2 = RR_{\text{rect}}$
- 6) Rotate each left camera point $x' \sim Hx$ where $H = KR_1$
- 7) Alter the focal length (inside K) to keep as much points within the original image size
- 8) Repeat 6 and 7 for right camera points using R_2

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Qui i passi in dettaglio maggiore

Ci serve solo e in quanto è lo spostamento della sola left visto che la right già l'allineiamo



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Epipolar Geometry

Question Time!



Ok altro pacco finito!